

Continuum Coulomb hardness with binary nuclear charges

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Abstract

We prove that approximating the electronic Coulomb spectral infimum in three-dimensional space is QMA-hard when positive integer nuclear charges are encoded in binary. The nuclei have distinct rational positions, the electron number is unary, and the energy is minimized over all antisymmetric continuum states and spin sectors. A deterministic classical polynomial-time reduction produces instances with threshold separation at least one. The nuclear charges may be exponentially large, but every output has polynomial bit length.

1 The problem and the result

The exact molecular electronic Hamiltonian has unusually rigid input data: its external potential is generated by a list of positive nuclei. A hardness proof for a matrix obtained by projecting onto supplied orbitals does not by itself establish hardness for this operator. Projection provides an upper bound on the ground energy, while a reduction also needs a lower bound that excludes all omitted continuum states. We give such a reduction for the integer-charge, rational-position encoding below.

1.1 The precise promise problem

Fix a positive integer c . An input consists of distinct positions $R_1, \dots, R_M \in \mathbb{Q}^3$, positive integers $Z_1, \dots, Z_M$, an electron number $N \geq 1$, and rational thresholds $a < b$. Here $M \geq 1$, N is encoded in unary, and the charges are encoded in binary. Every coordinate and threshold is encoded as a signed binary numerator and a positive binary denominator in lowest terms, using a fixed standard self-delimiting encoding. Write L for the total bit length and require $b - a \geq L^{-c}$.

On $\bigwedge^N L^2(\mathbb{R}^3; \mathbb{C}^2)$ consider, in atomic units,

$$H(R, Z, N) = -\frac{1}{2} \sum_{\ell=1}^N \Delta_{x_\ell} - \sum_{\ell=1}^N \sum_{\alpha=1}^M \frac{Z_\alpha}{|x_\ell - R_\alpha|} + \sum_{1 \leq \ell < k \leq N} \frac{1}{|x_\ell - x_k|}. \quad (1.1)$$

The form domain is

$$\mathcal{Q}_N = H^1((\mathbb{R}^3)^N; (\mathbb{C}^2)^{\otimes N}) \cap \bigwedge^N L^2(\mathbb{R}^3; \mathbb{C}^2).$$

Antisymmetry exchanges spatial and spin coordinates together. We use the lower-bounded self-adjoint Coulomb operator associated with the closed form on $\mathcal{Q}_N$. The three-dimensional Hardy inequality, applied with other coordinates fixed, gives infinitesimal form bounds for the finitely many Coulomb terms and justifies this convention. Set

$$E_0(R, Z, N) = \inf \text{Spec } H(R, Z, N).$$

The YES promise is $E_0 \leq a$, and the NO promise is $E_0 \geq b$. No requirement is made outside the promise.

There is no supplied basis, electron state, magnetic field, or independently adjustable external potential. The minimization includes every total-spin sector. Nuclear repulsion is omitted because the energy convention is electronic and the nuclei are clamped. The problem places no neutrality, fixed-element, or separate nuclear-separation promise on its inputs. In particular, the threshold gap is a precision requirement; it does not assert a molecular spectral gap, a discrete eigenvalue at the infimum, or an efficiently preparable ground state.

Theorem 1.1. *The promise problem above is QMA-hard under deterministic classical polynomial-time many-one reductions for $c = 1$. The reduction can produce instances with $b - a \geq 1$.*

Thus the exact clamped-nuclei problem is QMA-hard for a fixed inverse-polynomial precision, with the input conventions in Table 1. The binary encoding of unrestricted positive nuclear charges is essential to the construction given here. Charges and inverse nuclear separations may have exponentially large values; their descriptions still have polynomial length. The theorem makes no claim of membership in QMA, or of hardness under an additional bound on nuclear charges.

For clarity, QMA-hardness means that every promise problem admitting polynomial-time quantum verification with a polynomial-size quantum witness reduces to this one by a deterministic classical polynomial-time map preserving YES and NO instances. We use a standard QMA-complete Heisenberg problem as the source. The sole complexity-theoretic input is stated precisely in Theorem 2.1; the subsequent construction and its analytic estimates are proved below.

1.2 Context and the continuum obstruction

Electronic-structure complexity has several distinct formulations. Liu, Christandl, and Verstraete established QMA-completeness of the finite-mode N -representability problem under polynomial-time Turing reductions, using general two-body fermionic Hamiltonians in the argument [5]. Restricting the coefficients to those produced by an actual Coulomb potential is a further requirement. Whitfield, Love, and Aspuru-Guzik explained these distinctions and the role of magnetic fields in the early hardness results [16, Sections 3.1 and 6.1].

Schuch and Verstraete proved QMA-completeness for a Hubbard model with local magnetic fields and developed a continuum realization using designed scalar and spin-dependent potentials [13]. Their Hubbard-to-Heisenberg route and comparison with the complementary one-particle

Input or convention	Specification
Nuclear positions	Pairwise distinct $R_\alpha \in \mathbb{Q}^3$; binary rational coordinates
Nuclear charges	Positive integers Z_α , encoded in binary
Electron number	$N \geq 1$, encoded in unary
State space	Full spinful antisymmetric continuum space; all spin sectors
External data	Nuclear list only; no supplied orbital basis or magnetic field
Energy thresholds	Rational $a < b$; YES: $E_0 \leq a$; NO: $E_0 \geq b$
Promise precision	$b - a \geq L^{-1}$ in the theorem, for total input length L
Constructed separation	$b - a \geq 1$; a threshold separation, not a spectral gap
Energy convention	Clamped-nucleus electronic energy; nuclear repulsion omitted

Table 1: The input model and the precision attained by the reduction. Rational data use the self-delimiting encoding specified in the text.

space are relevant methodological predecessors. Positive point nuclei impose a different restriction on the external field; the continuum comparison needed here is proved in [Section 5](#).

O’Gorman, Irani, Whitfield, and Fefferman proved QMA-completeness for electronic structure in a supplied finite basis [[8](#), Theorem 1]. Their published Appendix D, Corollary 1, includes positive unit nuclear charges, but the basis remains part of the input; Section V explicitly raises the complete-space question. Indeed, if P is an orbital projection, the variational inequality

$$\inf \text{Spec } H \leq \min \text{Spec}(PHP|_{\text{Ran } P})$$

does not transfer a projected NO-instance lower bound to H . A continuum hardness proof must exclude lower states outside that projection.

The Heisenberg hardness theorem of Cubitt, Montanaro, and Piddock [[4](#), [3](#)] supplies a field-free square-lattice source with signed polynomially bounded interactions. Its efficient spatially sparse construction is relevant: generic simulation of a dense interaction graph would not by itself give the required polynomial interaction scales. We use the singlet mediator of Piddock and Montanaro [[11](#)] to obtain a positive Heisenberg model, and then use repulsive Hubbard superexchange [[1](#)]. The finite calculations below supply explicit scalar shifts, errors, rationalization, and the geometric control required by the nuclear construction.

Schleich et al. [[12](#)] study chemically motivated preparation and dynamics through assumptions on accessible chemical processes. Those assumptions define a different computational task from deciding every promised spectral-infimum instance considered here.

The companion article *QMA-hardness of continuum Coulomb energy with unit nuclear charges* [[9](#), Theorem 1] proves hardness with every $Z_\alpha = 1$. Its construction uses positive continuous charge density, polynomial confinement, and equal-mass cubature. That theorem implies hardness of the larger binary-charge input class, but it does not use the binary-charge theorem proved here. Conversely, splitting our exponentially large charges into individual unit nuclei would not give polynomial output size. The present proof retains a distinct quantitative contribution: certified exponential-accuracy eigenvalue computation and relative tunneling calibration for widely separated hydrogenic wells.

Binding questions are also distinct from the decision problem. For the same kinetic normalization, the companion ionization theorem [[10](#), Theorem 1.1] states that strict binding $E_N < E_{N-1}$ implies $N \leq Z_{\text{tot}} + CM$ for a universal C , where $Z_{\text{tot}} = \sum_\alpha Z_\alpha$ and $E_0 = 0$ in that particle-number notation. It assumes distinct nuclei and charges at least one, with no separation condition. Our reduction imposes no strict-binding promise, so that bound supplies neither the continuum exclusion estimate nor a ground-state existence assertion. It is a scope comparison only.

1.3 Mechanism of the reduction

We first replace each signed Heisenberg bond by positive interactions with a strongly bound ancillary singlet. A repulsive Hubbard model at half filling then realizes these positive exchanges through virtual double occupancy: its low-energy singly occupied states carry the spins, and the effective exchange is proportional to the square of the hopping strength. The continuum construction must therefore realize prescribed small positive hoppings together with a fixed onsite repulsion. The continuum comparison retains all configurations and spin sectors in the selected orbitals; single occupancy is obtained in the finite Hubbard analysis, not imposed on admissible continuum states.

The construction uses a large integer scale $Z_* \asymp e^D$, where D is a fixed polynomial in the source size. In dilated coordinates $y = Z_*x$, each site contains a nucleus of charge close to Z_* and

a weaker positive secondary nucleus. Their normalized one-electron states are close to hydrogen ground states. A secondary displacement tunes each isolated-well energy of the dilated operator to $-1/2$ with exponentially small error. The effective model uses the shifted physical energy divided by Z_* . In these units, hopping amplitudes are Z_* times their dilated one-electron values, while the onsite electron repulsion approaches the fixed constant $5/8$.

Three quantitative ingredients make the continuum and bit-complexity requirements compatible.

- (i) The sites are placed so that small displacements can independently change every required bond length to first order, with a quantitative bound on the displacement size. A relative tail estimate, proved with capped remote fields and comparison barriers, converts these changes into controlled hopping ratios. Absolute proximity to a hydrogen orbital would not give the required relative precision.
- (ii) A graded polynomial Ritz space evaluates a one-well eigenvalue to exponential accuracy in polynomial time. Uniform analytic estimates, an explicit rational mass-matrix lower bound, and certified eigenvalue enclosures make the procedure valid even for skinny cells and ill-conditioned intermediate bases. The unknown eigenfunction is used only to prove approximation, not as an input to the algorithm.
- (iii) Localization yields a fixed gap on the entire one-particle complement. A kinetic-trace and Hardy estimate controls Coulomb coupling on every fermionic superposition in the selected modes. Completing a square against the complementary gap gives a lower bound for the full many-electron spectral infimum.

These arguments provide quantitative tools for constructions involving exponentially weak tunneling and a continuum representation. Their uniformity is part of the proof: errors must remain small after multiplication by Z_* , and all required precision must cost only polynomially many bits. The numerical construction combines local polynomial approximation through a partition of unity, in the sense of Melenk and Babuška [7], with separate rational conditioning and integration bounds. The complement estimate uses spatial localization [14] and a quadratic elimination argument whose finite-dimensional analogue is second-order perturbation theory [2]. These methods are credited at their points of use; their needed quantitative forms are proved here.

The order of construction and the order of justification deserve separate attention. We first prescribe the finite Hubbard hoppings and compute the primary centers from positive integrals of explicit capped Coulomb fields. These surrogate fields do not involve any unknown orbital or secondary displacement. Only after rounding the primary centers and fixing the integer charges do we tune the secondary nuclei. The surrogate replacement bounds are uniform throughout the tuning cube, and the relative orbital estimates apply to every tuned output. Finally, the continuum lower bound transfers the Hubbard NO promise to arbitrary wavefunctions, and an affine change of energy transfers both promises to rational physical thresholds.

Section 2 develops the finite models and the contact layout. Section 3 proves the isolated-well estimates and the eigenvalue oracle. Section 4 computes the positions and calibrates hopping. Section 5 proves the full continuum comparison. Section 6 chooses all parameters, gives the classical reduction in its actual order of execution, and verifies the final thresholds and encoding.

Throughout, constants denoted by $C, c > 0$ are independent of the large scale parameters and may change between estimates. A polynomial bound always means a fixed polynomial with uniform coefficients and degree. Exponentially small bounds are stated explicitly when they are later amplified. All operator inequalities involving unbounded Hamiltonians are quadratic-form inequalities unless their bounded compressions are specified.

2 Finite models and a controllable contact geometry

We first reduce a signed spin Hamiltonian to a half-filled Hubbard model. The geometric construction is part of this reduction: every Hubbard hopping will later be set by a small displacement of a nuclear site. For a finite-dimensional self-adjoint operator A , write $E(A)$ for its smallest eigenvalue. On two spin-1/2 sites set

$$h_{ij}^s = X_i X_j + Y_i Y_j + Z_i Z_j = \sigma_i \cdot \sigma_j.$$

Its eigenvalues are -3 on the singlet and 1 on the triplet, so $\|h_{ij}^s\| = 3$.

2.1 The source problem and a quantitative perturbation bound

Theorem 2.1 (Source problem). *There is a QMA-complete promise problem whose Hamiltonians are*

$$H_{\text{in}} = \sum_{e=LR} J_e h_{LR}^s$$

on finite, nonwrapping subgraphs of the square lattice, with no one-qubit terms. The signed weights J_e are rational. The number of vertices and edges, the lattice extent, the weight magnitudes, and the threshold magnitudes are bounded by fixed polynomials in the source size $n \geq 2$. The rational thresholds $a_s < b_s$ have separation $\delta_s = b_s - a_s \geq 1/\text{poly}(n)$.

This is the square-lattice Heisenberg consequence of [3, Theorem 48 and Section 10.1], using the polynomial-weight spatially sparse branch of Lemma 47 there. The Heisenberg interaction has rank-three coupling matrix, so it is outside the exceptional rank-one family in that theorem; the square-lattice construction has no one-qubit fields. The following normalization makes the encoding interface explicit.

Let $\delta > 0$ be a known rational inverse-polynomial lower bound on the incoming gap, let q_s be the number of weighted edges, and put $\varepsilon_0 = \delta/100$. Round each edge coefficient within $\varepsilon_0/\max\{1, q_s\}$. Since $\|h_{ij}^s\| = 3$, the resulting Hamiltonian changes by norm at most $3\varepsilon_0$. If the source also has an identity coefficient c_0 and thresholds a_0, b_0 , approximate each of those three numbers within ε_0 , obtaining $\tilde{c}_0, \tilde{a}_0, \tilde{b}_0$. For the rounded Hamiltonian with its identity term removed, set

$$a_s = \tilde{a}_0 - \tilde{c}_0 + 5\varepsilon_0, \quad b_s = \tilde{b}_0 - \tilde{c}_0 - 5\varepsilon_0.$$

The two threshold approximation errors and the Hamiltonian error sum to at most $5\varepsilon_0$, so both promises survive. Moreover, $b_s - a_s \geq \delta - 12\varepsilon_0 > 0$. All rational descriptions have polynomial length. A connected grid component with v vertices has coordinate diameter at most $v - 1$, by following its lattice paths; translating components into disjoint strips gives polynomial extent. These are normalization steps applied to the cited source, not additional conclusions imported from its statement.

This imported source theorem is the only hardness theorem we need. In what follows zero-weight edges are omitted. No lower bound on the magnitudes of the remaining weights is assumed, and no total-spin sector is selected. By appending an isolated original vertex if necessary, at a lattice position outside the existing finite drawing, we may assume that there is at least one original vertex. Tensoring with its uncoupled spin preserves the minimum energy and all polynomial bounds.

We use the following finite-dimensional estimate twice. It controls the bottom energy and does not require a gap within the encoded subspace. The effective second-order operator is the familiar one from Schrieffer–Wolff perturbation theory; see Bravyi, DiVincenzo, and Loss [2, Section 3.2].

We prove the particular bound needed here directly, so its constants and the empty-complement case are explicit.

Lemma 2.2 (Second-order bottom-energy estimate). *Let $A \geq 0$ be a finite-dimensional self-adjoint operator, let $P \neq 0$ project onto its kernel, and put $Q = I - P$. Suppose $A \geq gQ$ for some $g > 0$. Let $B = B^*$ satisfy $PBP = 0$ and $\|B\| \leq \epsilon g$, where $0 \leq \epsilon < 1/2$. Define A^{-1} to vanish on P . Then*

$$\left| E(A + B) - E(-PBA^{-1}BP|_P) \right| \leq 2g\epsilon^3. \quad (2.1)$$

The conclusion also holds when $Q = 0$.

Proof. Let $K = PBA^{-1}BP \geq 0$ and $\kappa = \|K\|$. Since $Bp \in Q$ for $p \in P$,

$$0 \leq \kappa \leq \frac{\|B\|^2}{g} \leq g\epsilon^2. \quad (2.2)$$

For $p \in P$ and $q \in Q$, the inequality $QBQ \geq -\epsilon gQ \geq -\epsilon A|_Q$ gives

$$\begin{aligned} \langle p + q, (A + B)(p + q) \rangle &\geq (1 - \epsilon) \langle q, Aq \rangle + 2 \operatorname{Re} \langle q, Bp \rangle \\ &= (1 - \epsilon) \left\| A^{1/2}q + \frac{A^{-1/2}Bp}{1 - \epsilon} \right\|^2 - \frac{\langle p, Kp \rangle}{1 - \epsilon}. \end{aligned}$$

On a unit vector this is at least $-\kappa/(1 - \epsilon)$. Consequently

$$E(A + B) + \kappa \geq -\frac{\epsilon\kappa}{1 - \epsilon} \geq -2g\epsilon^3.$$

For the other direction choose a unit $p \in P$ with $Kp = \kappa p$ and set $q = -A^{-1}Bp$. Then $\|q\| \leq \epsilon$ and

$$\langle p + q, (A + B)(p + q) \rangle = -\kappa + \langle q, Bq \rangle.$$

Writing $r = \|q\|^2$, the variational principle therefore gives

$$E(A + B) + \kappa \leq \frac{\langle q, Bq \rangle + \kappa r}{1 + r} \leq g\epsilon^3 + g\epsilon^4 \leq \frac{3}{2}g\epsilon^3.$$

The smallest eigenvalue of $-K$ on P is $-\kappa$, proving the claim. If $Q = 0$, the hypothesis $PBP = 0$ forces $B = 0$, and the same formulas hold with all inverse terms zero. $\square$

2.2 Replacing signed bonds by positive Heisenberg bonds

Piddock and Montanaro's singlet mediator construction changes the sign of an effective interaction by coupling to the same or to opposite members of a strongly bound pair [11, Section 2.4 and Lemma 5]. We use its isotropic specialization below, keeping the scalar shift and the simultaneous-gadget error explicit. This stage produces only positive spin bonds; the next geometric stage assigns each of them a controllable unit contact.

For every source edge $e = LR$ introduce two ancillary spins A_e, B_e . There are k such pairs. Define

$$\begin{aligned} v_e &= \sqrt{2|J_e|}, \quad C_e = \begin{cases} A_e, & J_e < 0, \\ B_e, & J_e > 0, \end{cases} \\ H_+ &= \Delta \sum_e h_{A_e B_e}^s + \sqrt{\Delta} \sum_e v_e (h_{LA_e}^s + h_{RC_e}^s). \end{aligned} \quad (2.3)$$

Every nonzero coefficient in H_+ is positive. Put

$$V = \sum_e v_e, \quad C_0 = 3k\Delta + \frac{3}{2} \sum_e v_e^2. \quad (2.4)$$

Proposition 2.3 (Positive-spin reduction). *If $\sqrt{\Delta} > 3V$, then*

$$|E(H_+) - E(H_{\text{in}}) + C_0| \leq \frac{27V^3}{\sqrt{\Delta}}. \quad (2.5)$$

In particular, a polynomially large integer Δ makes this error an arbitrarily small prescribed inverse polynomial.

Proof. Take

$$A = \Delta \sum_e (h_{A_e B_e}^s + 3I), \quad B = \sqrt{\Delta} \sum_e v_e (h_{L A_e}^s + h_{R C_e}^s).$$

The kernel of A consists of arbitrary original spins tensored with a singlet on each ancillary pair. Denote its projection by P_s . The gap is $g = 4\Delta$, and

$$P_s B P_s = 0, \quad \|B\| \leq 6\sqrt{\Delta}V.$$

A weak interaction from pair e takes that pair's singlet into its triplet space and leaves every other ancillary pair in its singlet. Thus every state reached from P_s in one hop has excitation energy 4Δ . Moreover, terms associated with distinct pairs give zero mixed second-order contributions: a term acting on pair $f \neq e$ cannot remove the triplet on pair e . This remains true when the two source edges share an original vertex.

For a singlet $|s\rangle$ on A, B and Pauli components μ, ν ,

$$\langle s | \sigma_A^\mu \sigma_A^\nu | s \rangle = \delta_{\mu\nu}, \quad \langle s | \sigma_A^\mu \sigma_B^\nu | s \rangle = -\delta_{\mu\nu}.$$

Consequently the squares of the two weak terms each compress to $3I$. Their two ordered cross terms sum to $2h_{LR}^s$ when $C_e = A_e$, and to $-2h_{LR}^s$ when $C_e = B_e$. The contribution of pair e to $-P_s B A^{-1} B P_s$, identified with an operator on the original spins, is therefore

$$-\frac{v_e^2}{4} \begin{cases} 6I + 2h_{LR}^s, & J_e < 0, \\ 6I - 2h_{LR}^s, & J_e > 0 \end{cases} = J_e h_{LR}^s - \frac{3}{2} v_e^2 I. \quad (2.6)$$

The factor 4 in the denominator is the singlet–triplet excitation energy in units of Δ .

Apply Lemma 2.2 with $\epsilon = 3V/(2\sqrt{\Delta}) < 1/2$. Its error bound is $2(4\Delta)\epsilon^3 = 27V^3/\sqrt{\Delta}$. Since $A + B = H_+ + 3k\Delta I$, summing (2.6) gives the result. If $k = 0$, both perturbation and shift vanish and the assertion is exact. $\square$

Here is one explicit scale choice, which we retain for later bookkeeping. Set

$$\begin{aligned} \gamma &= \min\{1, \delta_s\}, & W &= \max(\{1\} \cup \{|J_e| : e\}), \\ \bar{V} &= \max\{1, 2kW\}, & \Delta &= \left\lceil \left(\frac{1728\bar{V}^3}{\gamma} \right)^2 \right\rceil. \end{aligned} \quad (2.7)$$

Then $V \leq \bar{V}$, and (2.5) is at most $\gamma/64$. Furthermore $v_e \leq \sqrt{\Delta}$, so every coefficient of H_+ is at most Δ . All these scales have polynomial magnitude in n . Intermediate square roots cause no encoding issue: their nonnegative values admit polynomial-precision rational approximations by bisection.

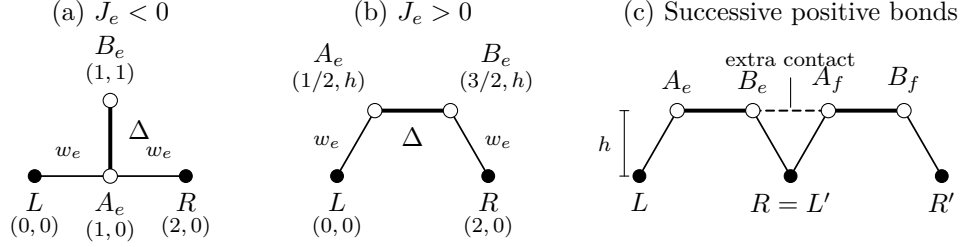


Figure 1: The contact construction in the local (ℓ, ζ) plane, with $h = \sqrt{3}/2$ and $w_e = \sqrt{\Delta} v_e$. Filled vertices are original spins and open vertices are ancillas. Every solid segment has length one; thick segments carry weight Δ and thin segments the indicated weak weight. The dashed segment in (c) is an additional unit contact, with weight zero in H_+ ; it receives only the small positive hopping floor in the Hubbard model. Horizontal gadgets use $\zeta = z$, and vertical gadgets use $\zeta = -z$.

2.3 Contact geometry and independent length control

Place the original lattice vertices at spacing 2 in the x, y plane. Orient each source edge in the positive x or positive y direction. Use local coordinates (ℓ, ζ) , where ℓ is distance from its left endpoint along that direction, and $\zeta = z$ for a horizontal edge but $\zeta = -z$ for a vertical edge. The two ancillas are placed at

	A_e	B_e
$J_e < 0$	$(1, 0)$	$(1, 1)$
$J_e > 0$	$(1/2, \sqrt{3}/2)$	$(3/2, \sqrt{3}/2)$

(2.8)

The local endpoints are always $L = (0, 0)$ and $R = (2, 0)$. Let $p_1, \dots, p_m$ denote all original and ancillary sites. Their coordinates belong to $\mathbb{Q}(\sqrt{3})^3$ and have polynomial bit descriptions. Figure 1 shows these placements and the additional contact between consecutive positive gadgets.

Lemma 2.4 (Contact classification). *All the sites p_i are distinct. Every pair is either at distance one or at distance at least $\sqrt{2}$. The complete set $\mathcal{C}$ of unit-distance pairs consists of:*

1. LA_e , A_eR , and A_eB_e for a negative source edge;
2. LA_e , A_eB_e , and B_eR for a positive source edge;
3. B_eA_f for two consecutive positive source edges e, f on the same original lattice line.

Thus every interaction of H_+ is a contact. Its weight on a contact will be denoted $K_{ij} \geq 0$, using $K_{ij} = 0$ for contacts in the third class.

Proof. Within one gadget the assertion follows directly from (2.8). For a negative gadget the two distances from B_e to original endpoints are $\sqrt{2}$; for a positive gadget the two nonadjacent endpoint-ancilla distances are $\sqrt{3}$. Original vertices are mutually separated by at least 2. An original vertex other than the endpoints of an ancilla's gadget, on that same axis line, has along-line distance at least $3/2$ to a positive ancilla and at least 3 to a negative midpoint. An original vertex off that line differs in a transverse coordinate by at least 2.

For ancillas belonging to distinct gadgets on the same axis line, the only along-line separation below $3/2$ is the separation 1 between B_e and A_f for consecutive positive gadgets. They have the same height and hence are a contact. All remaining such pairs are at least $3/2 > \sqrt{2}$ apart. Distinct parallel axis lines have transverse separation at least 2.

It remains to compare horizontal and vertical ancillas. Call an ancilla *midpoint type* if it belongs to a negative gadget, and *positive type* otherwise. Two midpoint-type projections differ by at least 1 in each planar coordinate, so their distance is at least $\sqrt{2}$. A midpoint-type and a positive-type projection differ by at least 1 and $1/2$ in the two planar coordinates. Their heights differ by at least $\sqrt{3}/2$, since horizontal and vertical heights have opposite signs. Their squared distance is therefore at least $1 + 1/4 + 3/4 = 2$. Finally two positive-type ancillas from perpendicular directions have height difference $\sqrt{3}$, already larger than $\sqrt{2}$. These cases exhaust all pairs and also establish distinctness. $\square$

Orient each contact arbitrarily and write $u_{ij} = p_j - p_i$, so $|u_{ij}| = 1$. Its first-order length change under site displacements s_i is $u_{ij} \cdot (s_j - s_i)$. Reversing the orientation reverses both factors and leaves this scalar unchanged. The following stronger-than-counting fact is what permits independent hopping calibration.

Lemma 2.5 (A bounded right inverse for contact lengths). *For every real assignment $(d_{ij})_{\{i,j\} \in \mathcal{C}}$ there are displacements $s_i \in \mathbb{R}^3$ such that*

$$u_{ij} \cdot (s_j - s_i) = d_{ij}, \quad \max_i |s_i| \leq 6m \max_{\mathcal{C}} |d_{ij}|. \quad (2.9)$$

For an empty contact set take all displacements zero. The construction is linear in d , uses polynomially many arithmetic operations with coefficients in $\mathbb{Q}(\sqrt{3})$, and has no consistency condition around lattice cycles.

Proof. Set the height displacement of every original vertex to zero. The horizontal coordinate of each original vertex will be determined using only its original row, and its vertical planar coordinate using only its original column. These are independent variables even when both directions meet at the same vertex.

Fix one such axis line. Write a for displacement along it, z for physical height displacement, and put $\sigma = 1$ for a horizontal line and $\sigma = -1$ for a vertical line. On every connected run of negative bonds choose the first original a coordinate to be zero, and accumulate

$$\begin{aligned} a_R &= a_L + d_{LA_e} + d_{A_e R}, & a_{A_e} &= a_L + d_{LA_e}, \\ a_{B_e} &= a_{A_e}, & z_{A_e} &= 0, & z_{B_e} &= \sigma d_{A_e B_e}. \end{aligned} \quad (2.10)$$

Unconstrained original coordinates may be set to zero. A positive or missing bond places no requirement on the difference of its original endpoints' a coordinates at this step, so it separates runs. Each run is a path on a line, and (2.10) satisfies all three constraints in its negative gadgets.

Now consider the elevated vertices of positive gadgets on this line. Their same-height contacts, including the extra contacts of Lemma 2.4, form disjoint paths. Set the first a coordinate on each path to zero and successively set $a_{\text{next}} = a_{\text{current}} + d_{\text{edge}}$. Every such contact vector is along the axis, so its constraint involves only these a coordinates. Each elevated vertex has exactly one additional, sloped contact to an original vertex. Writing $h_0 = \sqrt{3}/2$, set

$$\begin{aligned} z_{A_e} &= \frac{d_{LA_e} - (a_{A_e} - a_L)/2}{\sigma h_0}, \\ z_{B_e} &= \frac{d_{B_e R} - (a_R - a_{B_e})/2}{\sigma h_0}. \end{aligned} \quad (2.11)$$

Indeed the two unit spoke vectors are $\frac{1}{2}e_{\text{axis}} + \sigma h_0 e_z$ and $\frac{1}{2}e_{\text{axis}} - \sigma h_0 e_z$, respectively. Because the original height displacements vanish, these formulas satisfy the remaining equations without

imposing any new condition on original coordinates. Set each ancilla's unused transverse coordinate to zero.

Repeat on all rows and columns. The row procedure changes only original x coordinates, and the column procedure only original y coordinates. No ancilla belongs to both procedures. Thus the constructions do not overwrite each other. In particular a square cycle is not a cycle in either scalar accumulation graph; its vector closure is automatic because actual vertex displacements have been assigned.

For the norm bound put $d_{\max} = \max_{\mathcal{C}} |d_{ij}|$. Every original along-line coordinate has magnitude at most $2md_{\max}$, and every accumulated elevated coordinate has magnitude at most $md_{\max}$. Negative ancilla coordinates have magnitude at most $(2m+1)d_{\max}$ along the line and at most $d_{\max}$ in height. From (2.11), positive ancilla heights are at most $(1+3m/2)d_{\max}/h_0$. Combining at most two nonzero coordinates at any site gives the stated bound $6md_{\max}$. All assignments are additions and multiplication by fixed rational numbers or by $1/h_0$, proving the algorithmic assertion. $\square$

Only approximate length equations will be needed. To make their bit complexity explicit, suppose the data are known within $\tau/4$ and apply the displayed right inverse to those approximations. Approximate its output and Dp_i , then round to rational positions X_i , with total Euclidean error at most $\tau/4$ per site. Defining $s_i = X_i - Dp_i$ absorbs this rounding even though the p_i may be irrational. Every contact equation then has error at most $\tau/4 + 2(\tau/4) < \tau$, and

$$\max_i |s_i| \leq 6m(\max_{\mathcal{C}} |d_{ij}| + \tau) + \tau. \quad (2.12)$$

The number of arithmetic operations is polynomial, and all accumulated coefficients have polynomial size. Computing $\sqrt{3}$ and the other real inputs to sufficiently many additional polynomially bounded bits therefore gives the stated absolute accuracy in polynomial time. In particular this does not call for an ill-conditioned general linear-system solve.

For later use, actual distances satisfy the exact vector identities and the elementary estimates

$$\begin{aligned} X_j - X_i &= Du_{ij} + s_j - s_i & (\{i, j\} \in \mathcal{C}), \\ |X_j - X_i| &= D + u_{ij} \cdot (s_j - s_i) + O(S^2/D) & (\{i, j\} \in \mathcal{C}), \\ |X_j - X_i| &\geq \sqrt{2}D - 2S & (\{i, j\} \notin \mathcal{C}), \end{aligned} \quad (2.13)$$

provided $\max_i |s_i| \leq S$ and $S \leq D/4$. The middle estimate is the Taylor bound for the Euclidean norm away from zero. The last is the triangle inequality and Lemma 2.4. No exact nonlinear prescribed-length problem is being solved; the quadratic remainder will be included in the hopping estimates.

2.4 The half-filled Hubbard model

We now realize those positive bonds as second-order hopping processes. Virtual double occupancy gives the antiferromagnetic superexchange mechanism of Anderson [1]. In our Pauli-matrix normalization the coefficient is t^2/U multiplying $h_{ij}^s - I$. We calculate the fermionic signs below and retain the constant term, because it contributes to the final decision thresholds.

Associate two fermionic modes, labelled $\uparrow, \downarrow$, to each of the m sites and restrict to exactly m fermions. Let $c_{i\sigma}^\dagger, c_{i\sigma}$ satisfy the canonical anticommutation relations and put $n_{i\sigma} = c_{i\sigma}^\dagger c_{i\sigma}$. For

nonnegative hopping strengths define

$$H_{\text{Hub}}(t) = - \sum_{\{i,j\} \in \mathcal{C}} \sum_{\sigma=\uparrow,\downarrow} t_{ij} (c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (2.14)$$

$$U = \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{\phi(y)^2 \phi(y')^2}{|y - y'|} dy dy', \quad \phi(y) = \pi^{-1/2} e^{-|y|}.$$

The radial probability density of ϕ^2 is $p(r) = 4r^2 e^{-2r}$. Angular integration gives the kernel $1/\max(r, s)$: for $r, s > 0$,

$$\frac{1}{2} \int_{-1}^1 \frac{du}{\sqrt{r^2 + s^2 - 2rsu}} = \frac{1}{\max(r, s)}.$$

Since $\int_0^r p(s) ds = 1 - e^{-2r}(1 + 2r + 2r^2)$,

$$U = 8 \int_0^\infty r e^{-2r} [1 - e^{-2r}(1 + 2r + 2r^2)] dr$$

$$= 8 \left(\frac{1}{4} - \frac{1}{16} - \frac{1}{16} - \frac{3}{64} \right) = \frac{5}{8}. \quad (2.15)$$

Thus U is a fixed positive rational constant.

Let $q_{\mathcal{C}} = |\mathcal{C}|$ and define the rational scales

$$\bar{R} = \max\{1, q_{\mathcal{C}} \Delta\}, \quad \eta = \frac{U\gamma}{1024\bar{R}^3}, \quad \alpha = \frac{\eta^2}{U}, \quad f = \frac{\alpha\gamma}{128 \max\{1, q_{\mathcal{C}}\}}. \quad (2.16)$$

The ideal hoppings are $t_{ij}^{(0)} = \eta \sqrt{K_{ij}}$. For each contact compute a rational r_{ij} with $|r_{ij} - t_{ij}^{(0)}| \leq f$, and output

$$t_{ij} = \max\{f, r_{ij}\}. \quad (2.17)$$

Polynomial-precision bisection on nonnegative square roots (at most two nested square roots here) suffices to compute r_{ij} ; no exact comparison between irrational numbers is required. The maximum in (2.17) is a comparison of rationals. All final hoppings satisfy $1/\text{poly}(n) \leq t_{ij} < 1$, including those with $K_{ij} = 0$.

Proposition 2.6 (Hubbard reduction with rational hoppings). *The construction above is deterministic and polynomial time. With*

$$C_s = C_0 + \sum_{\{i,j\} \in \mathcal{C}} K_{ij} = 3k\Delta + \frac{3}{2} \sum_e v_e^2 + \sum_{\{i,j\} \in \mathcal{C}} K_{ij}, \quad (2.18)$$

it satisfies

$$|E(H_{\text{Hub}}(t)) - \alpha(E(H_{\text{in}}) - C_s)| \leq \frac{3\alpha\gamma}{64}. \quad (2.19)$$

The scalar C_s has polynomial magnitude and can be approximated to any polynomial-bit absolute precision in polynomial time.

Proof. First use the ideal hoppings $t^{(0)}$. Let $A_U = U \sum_i n_{i\uparrow} n_{i\downarrow}$ and let T be the hopping part of $H_{\text{Hub}}(t^{(0)})$. At filling m , the kernel of A_U is exactly the singly occupied subspace: having m fermions, at most one at each of m sites, forces precisely one per site. Identify this subspace with m spins, and denote its projection by P_{so} . The complementary excitation energies are positive

integer multiples of U . A single hop from P_{so} creates one vacant site and one doubly occupied site and hence has energy exactly U . In particular $P_{\text{so}}TP_{\text{so}} = 0$.

A second hop can return to single occupancy only by moving an electron from that doubly occupied site to that vacant site. Thus the two hops must use the same unordered bond; products from different bonds have zero compression, even when the bonds share a vertex. For completeness, consider one bond i, j of strength t , with local mode order $i \uparrow, i \downarrow, j \uparrow, j \downarrow$. Write $|D_i\rangle = c_{i\uparrow}^\dagger c_{i\downarrow}^\dagger |0\rangle$ and similarly for $|D_j\rangle$. The bond hopping operator T_{ij} obeys

$$T_{ij}|\uparrow, \downarrow\rangle = -t(|D_i\rangle + |D_j\rangle), \quad T_{ij}|\downarrow, \uparrow\rangle = t(|D_i\rangle + |D_j\rangle).$$

It consequently annihilates all triplets and maps the singlet to $-\sqrt{2}t(|D_i\rangle + |D_j\rangle)$, of squared norm $4t^2$. In an ordering containing other sites between i and j , those sites are singly occupied during this two-hop process; their fixed fermion parity changes transition phases but not the compressed product. Equivalently, permuting the intervening mode pairs gives the same calculation under a fermionic change of basis. It follows that

$$-P_{\text{so}}T_{ij}A_U^{-1}T_{ij}P_{\text{so}} = -\frac{4t^2}{U}P_{\text{singlet},ij} = \frac{t^2}{U}(h_{ij}^s - I). \quad (2.20)$$

Here $P_{\text{singlet},ij} = (I - h_{ij}^s)/4$.

For each spin, $c_{i\sigma}^\dagger c_{j\sigma} + c_{j\sigma}^\dagger c_{i\sigma}$ has norm one, so

$$\|T\| \leq 2 \sum_{\mathcal{C}} t_{ij}^{(0)} = 2\eta \sum_{\mathcal{C}} \sqrt{K_{ij}} \leq 2\eta \bar{R}.$$

The last inequality uses $K_{ij} \leq \Delta$ and $\Delta \geq 1$. Lemma 2.2, with $g = U$ and $\epsilon = 2\eta \bar{R}/U < 1/2$, and (2.20) give

$$\left| E(H_{\text{Hub}}(t^{(0)})) - \alpha \left(E(H_+) - \sum_{\mathcal{C}} K_{ij} \right) \right| \leq \frac{16\eta^3 \bar{R}^3}{U^2} \leq \frac{\alpha\gamma}{64}. \quad (2.21)$$

If the complementary subspace is empty, as can happen for $m = 1$, there is no hopping and this assertion is exact.

For the rational hoppings, nonnegativity of $t_{ij}^{(0)}$ and the accuracy of r_{ij} imply $|\max\{f, r_{ij}\} - t_{ij}^{(0)}| \leq f$. Therefore

$$\|H_{\text{Hub}}(t) - H_{\text{Hub}}(t^{(0)})\| \leq 2qcf \leq \frac{\alpha\gamma}{64}. \quad (2.22)$$

Also $t_{ij}^{(0)} \leq \eta \bar{R} \leq U/1024$ and $f \leq U/(128 \cdot 1024^2)$, which verify $f \leq t_{ij} < 1$. The floor f is an inverse polynomial. This argument covers an ideal hopping much smaller than f and an ideal hopping equal to zero; no coefficient lower bound has been used. If $\mathcal{C}$ is empty, (2.22) is simply zero.

Finally, Proposition 2.3 with (2.7) contributes at most another $\alpha\gamma/64$. Adding the three errors proves (2.19). All sizes, magnitudes, and reciprocal precision budgets in the construction are polynomial in n . Each K_{ij} is either Δ , a square root of the rational number $2\Delta|J_e|$, or zero. The displayed scalar shift is thus efficiently approximable to the asserted precision, completing the proof. $\square$

3 Isolated wells and deterministic energy tuning

This section supplies an exponentially accurate classical procedure for tuning the individual well energies. The procedure operates on one-electron operators only; it does not diagonalize any of the spin or Hubbard Hamiltonians from [Section 2](#).

There are two separate tasks. First we must show that moving a secondary positive nucleus changes one isolated energy monotonically, with much weaker effects on the other sites. Then we must evaluate those energies accurately enough to implement the resulting contraction. The required accuracy is exponential in D : an inverse-polynomial energy error in these dilated units would grow after the final amplification. The analytic and arithmetic estimates below address that precision cost.

Fix centers $X_1, \dots, X_m \in \mathbb{R}^3$ satisfying

$$|X_i - X_j| \geq D - 2S \quad (i \neq j), \quad S \leq D/100. \quad (3.1)$$

Here D is a sufficiently large integer and m/D is smaller than a sufficiently small absolute constant. All constants in this section are uniform in the centers and the parameters $\rho_i \in [1, 4]$. The stronger scale requirements used later imply these hypotheses. Computational statements additionally assume rational centers. Their bit lengths, rather than their numerical magnitudes, enter the complexity bounds.

Choose an integer Z_* with $e^D/2 \leq Z_* \leq 2e^D$. Round the following two numbers to nearest integers, once and for all, before tuning:

$$\begin{aligned} Z_* q &= \text{round}\left(Z_* \left(1 - \frac{500m}{D}\right)\right), \\ Z_* q' &= \text{round}\left(Z_* \frac{1000m}{D}\right). \end{aligned} \quad (3.2)$$

Both integers are positive for the scales considered here. In particular,

$$\begin{aligned} q &= 1 - \frac{500m}{D} + O(Z_*^{-1}), \\ q' &= \frac{1000m}{D} + O(Z_*^{-1}), \quad c \frac{m}{D} \leq q' \leq C \frac{m}{D}. \end{aligned} \quad (3.3)$$

The primary and secondary nuclei of site i are at X_i and $X_i + \rho_i e_z$, respectively, in scaled coordinates. Write

$$\begin{aligned} r_i &= |y - X_i|, \quad V_i(y) = \frac{q}{r_i} + \frac{q'}{|y - X_i - \rho_i e_z|}, \\ h &= -\frac{1}{2}\Delta - \sum_{i=1}^m V_i. \end{aligned} \quad (3.4)$$

These are the one-electron operators obtained from physical coordinates by $y = Z_* x$ and division of energies by Z_*^2 .

Fix a nondecreasing C^2 function $\chi : [0, \infty) \rightarrow [0, 1]$ that vanishes on $[0, 1/16]$ and equals one on $[1/8, \infty)$. It can have rational piecewise-polynomial data: integrate a normalized positive multiple of $(t - 1/16)^2(1/8 - t)^2$ on the transition interval and extend constantly outside. Set

$$\chi_i(y) = \chi(r_i/D), \quad W_i = \sum_{k \neq i} \chi_k V_k, \quad h_i = -\frac{1}{2}\Delta - V_i - W_i. \quad (3.5)$$

The caps remove the other nuclear singularities from h_i while retaining their fields near site i .

3.1 Uniform ground-state estimates

We use the three-dimensional Sobolev bounds $\|u\|_6 \leq C\|u\|_{H^1}$ and $\|u\|_\infty \leq C\|u\|_{H^2}$, together with their localized versions. We also use the translated Hardy inequality

$$\left\| \frac{u}{|\cdot - a|} \right\|_2 \leq 2\|\nabla u\|_2 \quad (a \in \mathbb{R}^3). \quad (3.6)$$

Indeed, integrate $\operatorname{div}((y - a)/|y - a|^2) = |y - a|^{-2}$ against $|u|^2$ for a smooth compactly supported u . Integration by parts and Cauchy–Schwarz give

$$\|u/|\cdot - a|\|_2^2 \leq 2\|u/|\cdot - a|\|_2 \|\nabla u\|_2.$$

Approximation proves (3.6) on H^1 . Thus Coulomb multiplication maps H^1 continuously into L^2 , uniformly in the location of the pole.

Proposition 3.1 (Uniform isolated-well estimates). *The operators h_i are self-adjoint on $H^2(\mathbb{R}^3)$, with form domain $H^1(\mathbb{R}^3)$. Each has a simple ground eigenvalue λ_i and a real, positive, normalized eigenfunction ψ_i . There is a constant $g > 0$ such that the remaining spectrum is at least $\lambda_i + g$. For $\phi_i(y) = \pi^{-1/2}e^{-r_i}$,*

$$|\lambda_i + 1/2| \leq C\frac{m}{D}, \quad \|\psi_i - \phi_i\|_{H^2} \leq C\frac{m}{D}. \quad (3.7)$$

Moreover, for derivative orders $j = 0, 1, 2$,

$$\|D^j W_i\|_\infty \leq C\frac{m}{D^{j+1}}, \quad 0 \leq W_i \leq \frac{100m}{D}. \quad (3.8)$$

These assertions hold before tuning and uniformly throughout $[1, 4]^m$.

Proof. On the nonzero support of χ_k , $r_k \geq D/16$. Since $\rho_k \leq 4$,

$$|y - X_k - \rho_k e_z| \geq r_k - 4 \geq r_k/2$$

for large D . The secondary singularity is inside the region where the cap is identically zero. Differentiation through order two, including derivatives of the cap, gives $\|D^j(\chi_k V_k)\|_\infty \leq CD^{-j-1}$. Summing proves the derivative bounds. For example, $q \leq 1$, $q' \leq 1$, and sufficiently large D give a bound below $100/D$ for each capped potential, proving the stated function bound.

Put $H_i^0 = -\Delta/2 - r_i^{-1}$ and $B_i = h_i - H_i^0$. Hardy's inequality and (3.8) give

$$\|B_i u\|_2 \leq C\frac{m}{D}\|u\|_{H^1}. \quad (3.9)$$

For any potential in these single-well operators, Hardy and Fourier interpolation give, for each $a > 0$, a uniform bound $\|Vu\|_2 \leq a\|\Delta u\|_2 + C_a\|u\|_2$. The Kato–Rellich theorem for relatively bounded perturbations [15, Theorem 6.4, p. 135] therefore gives domain H^2 . The same estimates, with the Laplacian term absorbed, prove uniform graph-norm equivalence:

$$C^{-1}\|u\|_{H^2} \leq \|h_i u\|_2 + C\|u\|_2 \leq C\|u\|_{H^2}. \quad (3.10)$$

The corresponding statement holds for H_i^0 . At form level a fixed positive shift of either operator is bounded above and below by constant multiples of $\|u\|_{H^1}^2$, giving the asserted form domains.

The hydrogenic ground state and its isolation follow without an excited-state formula. Direct integration by parts gives

$$\langle u, (H_i^0 + 1/2)u \rangle = \frac{1}{2} \left\| \nabla u + \frac{y - X_i}{r_i} u \right\|_2^2. \quad (3.11)$$

Its zero vectors are exactly the multiples of ϕ_i . If normalized vectors orthogonal to ϕ_i had energies tending to $-1/2$, they would be bounded in H^1 . The Coulomb form is compact under weak convergence of such a sequence: within radius a of the pole its absolute value is bounded by $Ca\|u\|_{H^1}^2$, by Sobolev and Hölder; beyond radius R it is bounded by $R^{-1}\|u\|_2^2$; on the intervening annulus local compactness applies. A weak limit would therefore obey

$$\langle u, H_i^0 u \rangle \leq -1/2, \quad \langle u, H_i^0 u \rangle \geq -\frac{1}{2}\|u\|_2^2 \geq -1/2.$$

Equality forces norm one and a multiple of ϕ_i , contrary to orthogonality. This proves a fixed gap $g_0 > 0$.

By (3.9), after a fixed positive shift the perturbed form lies between $1 - Cm/D$ and $1 + Cm/D$ times the unperturbed shifted form. The variational principle puts its first level within Cm/D of $-1/2$, and its second variational level at least $-1/2 + g_0 - Cm/D$. The negative first level is attained. The preceding truncation argument gives form compactness of both own Coulomb potentials and of the bounded decaying W_i . A normalized minimizing sequence is bounded in H^1 . Passing to a weakly convergent subsequence with limit u , compactness and lower semicontinuity give

$$\lambda_i \|u\|_2^2 \leq \langle u, h_i u \rangle \leq \lambda_i < 0.$$

Since $\|u\|_2 \leq 1$, these inequalities force $\|u\|_2 = 1$ and attainment. For small enough m/D the first level is simple and separated from the rest by a fixed $g > 0$. Taking the absolute value of a real minimizer cannot increase its energy. The Euler equation and the strong maximum principle on the connected complement of the poles make it positive there.

The form bounds give $\|\psi_i\|_{H^1} \leq C$. Its equation implies

$$(H_i^0 + 1/2)\psi_i = (\lambda_i + 1/2)\psi_i - B_i\psi_i,$$

with right side of norm at most Cm/D . On $\phi_i^\perp$, the inverse of $H_i^0 + 1/2$ is bounded from L^2 to H^2 : use the gap for the L^2 bound, then (3.10). The orthogonal component of ψ_i consequently has H^2 norm $O(m/D)$. Normalization and the positive overlap with ϕ_i put its coefficient along ϕ_i within $O((m/D)^2)$ of one. This proves (3.7) and the uniform sup bound. Positivity also holds at the poles by continuity and this sup-norm approximation, since both poles lie within distance four of X_i . $\square$

3.2 Moving the secondary nucleus

The derivative of a moving Coulomb potential is not an L^2 -bounded multiplier. We instead control its expectations by Hardy's inequality.

Lemma 3.2 (Bracketing and uniform secant bounds). *With centers and rounded charges fixed, for every choice of the other parameters, $\lambda_i(\rho_i = 1) < -1/2 < \lambda_i(\rho_i = 4)$. For $1 \leq a < b \leq 4$, holding the other parameters fixed,*

$$cq'(b - a) \leq \lambda_i(b) - \lambda_i(a) \leq Cq'(b - a). \quad (3.12)$$

If instead ρ_i is held fixed, then

$$|\lambda_i(\rho) - \lambda_i(\tilde{\rho})| \leq Cq' \frac{m}{D^2} \|\rho - \tilde{\rho}\|_\infty. \quad (3.13)$$

Proof. Taking the inner product of the eigen-equation with ϕ_i gives

$$\lambda_i = -\frac{1}{2} + (1 - q) - q'f(\rho_i) - \langle \phi_i, W_i \phi_i \rangle + O((m/D)^2), \quad (3.14)$$

where

$$f(\rho) = \int_{\mathbb{R}^3} \frac{\pi^{-1} e^{-2|y|}}{|y - \rho e_z|} dy.$$

Here $\langle \phi_i, \psi_i \rangle = 1 + O((m/D)^2)$, and replacing ψ_i by ϕ_i in a perturbing expectation costs $O((m/D)^2)$, by (3.9) and (3.7). Also $\langle \phi_i, r_i^{-1} \phi_i \rangle = 1$. The spherical average of $|r\omega - \rho e_z|^{-1}$ is $1/\max(r, \rho)$, as direct integration in $\omega \cdot e_z$ shows. Consequently,

$$f(\rho) = \rho^{-1} \int_0^\rho 4r^2 e^{-2r} dr + \int_\rho^\infty 4r e^{-2r} dr, \quad (3.15)$$

$$f'(\rho) = -\rho^{-2} \int_0^\rho 4r^2 e^{-2r} dr. \quad (3.16)$$

In particular,

$$f(1) = 1 - 2e^{-2} > 0.7, \quad f(4) \leq \frac{1}{4}, \quad \frac{1 - 5e^{-2}}{16} \leq -f'(\rho) \leq 1 \quad (1 \leq \rho \leq 4).$$

At the first endpoint the leading shift in (3.14) is at most $(500 - 700)m/D$; at the second it is at least $(500 - 250 - 100)m/D$. The quadratic and rounding errors fit inside these strict margins, proving bracketing.

For the secant bound, fix a ground function u at either endpoint, and put $A(t) = X_i + te_z$. The kernel derivative is dominated by $|y - A(t)|^{-2}$, and Hardy gives

$$\begin{aligned} \int \frac{|u^2 - \phi_i^2|}{|y - A(t)|^2} dy &\leq \left\| \frac{u - \phi_i}{|\cdot - A(t)|} \right\|_2 \left\| \frac{u + \phi_i}{|\cdot - A(t)|} \right\|_2 \\ &\leq 4 \|\nabla(u - \phi_i)\|_2 (\|\nabla u\|_2 + \|\nabla \phi_i\|_2) \leq C \frac{m}{D}. \end{aligned} \quad (3.17)$$

The same bound with u^2 justifies absolute integration in y and t . Thus replacing the density by ϕ_i^2 when integrating the moving-pole kernel derivative incurs at most $Cq'(m/D)(b - a)$ error. The variational principle gives

$$\langle \psi_i(b), (h_i(b) - h_i(a)) \psi_i(b) \rangle \leq \lambda_i(b) - \lambda_i(a) \leq \langle \psi_i(a), (h_i(b) - h_i(a)) \psi_i(a) \rangle.$$

Both bounding quantities differ from $-q'(f(b) - f(a))$ by the error just estimated, proving (3.12).

For $k \neq i$, the moving secondary pole in $\chi_k V_k$ stays at least $D/32$ from the region where its cap is nonzero. Its bounded potential changes in sup norm by at most $Cq'D^{-2}|\rho_k - \tilde{\rho}_k|$. Summing and applying the variational principle for a bounded additive perturbation proves (3.13). $\square$

3.3 Uniform tails

Lemma 3.3 (Crude tails before tuning). *Uniformly for all $\rho \in [1, 4]^m$,*

$$|\psi_i(y)| \leq C \exp[-.99(r_i - .01D)], \quad (3.18)$$

and, for sufficiently large D ,

$$\|\psi_i\|_{H^1(\{r_i > .65D\})} \leq e^{-.61D}. \quad (3.19)$$

Proof. On $r_i \geq .01D$, both own poles are far away and $V_i + W_i = O(m/D)$. For $a = .99$ and $b(y) = C \exp[-a(r_i - .01D)]$, direct calculation gives

$$\frac{(h_i - \lambda_i)b}{b} = -\frac{a^2}{2} + \frac{a}{r_i} - V_i - W_i - \lambda_i.$$

Its constant part is $(1 - a^2)/2 + O(m/D) > 0$ at the chosen scales. The zero-order coefficient of $h_i - \lambda_i$ is also positive throughout this exterior region. The uniform sup bound fixes C so that $b \geq |\psi_i|$ at the inner boundary. An $H^2(\mathbb{R}^3)$ function is uniformly continuous and vanishes at infinity; otherwise uniform continuity on a sequence of disjoint balls would contradict its square integrability. On a large finite annulus add a constant $\varepsilon > 0$ to b to dominate both signs of ψ_i at the outer boundary. The supersolution inequality is preserved. Testing the negative part of $b + \varepsilon \mp \psi_i$ proves comparison. Send the outer radius to infinity and then ε to zero. Inside $.01D$, the same sup bound proves (3.18).

For the gradient estimate choose a smooth cutoff ζ that vanishes on $r_i \leq .645D$, equals one on $r_i \geq .65D$, and has gradient $O(D^{-1})$. Testing the eigen-equation with $\zeta^2 \psi_i$ gives

$$\frac{1}{2} \|\nabla(\zeta \psi_i)\|_2^2 = \int (V_i + W_i + \lambda_i) |\zeta \psi_i|^2 dy + \frac{1}{2} \int |\nabla \zeta|^2 |\psi_i|^2 dy.$$

All coefficients on this support are bounded. Thus (3.18) bounds the exterior H^1 norm by a polynomial in D times $\exp[-.99(.645 - .01)D]$. Since $.99(.645 - .01) = .62865 > .61$, the polynomial factor is absorbed for large D . $\square$

3.4 An analytic estimate with uniform constants

We give the local regularity argument used in the numerical procedure. It explains why the shrinking length scales near a moving Coulomb pole do not increase the required polynomial degree beyond $O(D)$.

Lemma 3.4 (Uniform analytic patches). *Suppose u is bounded on the unit ball in $\mathbb{R}^3$ and satisfies $\Delta u = Fu$ there, where*

$$\|\partial^\beta F\|_\infty \leq A^{|\beta|+1} |\beta|! \quad \text{for every multi-index } \beta. \quad (3.20)$$

There is a constant A_1 , depending only on A , such that

$$\max_{|\alpha|=k} \|\partial^\alpha u\|_{L^\infty(B_{1/4})} \leq \|u\|_{L^2(B_1)} A_1^{k+1} k!. \quad (3.21)$$

On a sufficiently small fixed ball about the origin, Taylor polynomials consequently have geometrically decreasing errors in both value and first derivative, with constants depending only on A .

Proof. For two concentric balls of radii between $1/2$ and 1 , separated by h , the local estimates needed below are

$$\|D^2v\|_{2,\text{inner}} \leq C(\|\Delta v\|_{2,\text{outer}} + h^{-2}\|v\|_{2,\text{outer}}), \quad (3.22)$$

$$\|\nabla v\|_{2,\text{inner}} \leq C(h\|\Delta v\|_{2,\text{outer}} + h^{-1}\|v\|_{2,\text{outer}}). \quad (3.23)$$

To obtain them, insert smooth cutoffs with first and second derivatives $O(h^{-1})$ and $O(h^{-2})$. Integration by parts against a squared cutoff and Cauchy–Schwarz, with $ab \leq (ha)^2/2 + (b/h)^2/2$, bound first derivatives on an intermediate ball as in (3.23). Apply the whole-space Fourier inequality $\|D^2w\|_2 \leq C\|\Delta w\|_2$ to the cutoff function and substitute that first-derivative bound in its commutator terms to obtain (3.22). Approximation proves these estimates for weak solutions. Successive localized Laplacian regularity makes u smooth when F is smooth, justifying the following differentiations.

Fix an integer $\ell \geq 2$, set $r_j = 1 - j/(2\ell)$, and write

$$T_j = \max_{|\alpha|=j} \|\partial^\alpha u\|_{L^2(B_{r_j})}, \quad M = \|u\|_{L^2(B_1)}.$$

The first-derivative bound gives $T_1 \leq C(\ell + A/\ell)M$. For $j \leq \ell - 2$, apply (3.22) to an order- j derivative between B_{r_j} and $B_{r_{j+2}}$. Leibniz' rule yields

$$T_{j+2} \leq C \left(\sum_{b=0}^j \binom{j}{b} A^{b+1} b! T_{j-b} + \ell^2 T_j \right). \quad (3.24)$$

Lower-order derivatives are available on the larger balls $B_{r_{j-b}}$. The multi-index coefficient sum of order b is $\binom{j}{b}$, by the multinomial identity. If preceding orders satisfy $T_s \leq M(K\ell)^s$, the sum is at most

$$AM(K\ell)^j \sum_{b=0}^j (A/K)^b \leq 2AM(K\ell)^j$$

for $K \geq 2A$, using $\binom{j}{b} b! \leq \ell^b$. Increasing the fixed K absorbs the remaining $C\ell^2$ factor into $(K\ell)^2$. Both parity chains therefore satisfy $T_j \leq M(K\ell)^j$.

Set $\ell = k + 2$. Sobolev embedding on fixed smaller balls bounds the sup norm of an order- k derivative on $B_{1/4}$ by $CM[K(k+2)]^{k+2}$. The inequality $k! \geq (k/e)^k$, with the small cases absorbed in the constant, converts this into (3.21). Taylor's formula, summed over multi-indices, bounds the degree- L remainder on $|z| \leq a$ by a constant times $(C_1 a)^{L+1}$. The first-derivative remainder has an additional factor polynomial in L and the same geometric decay. Fixing a small enough that $C_1 a < 1$ proves the last assertion. $\square$

3.5 A polynomial-size graded Ritz space

Graded meshes with increasing polynomial degree are a standard way to combine singular local behavior with exponential approximation away from the singularities. For related eigenvalue approximation results with singular potentials, see Maday and Marcati [6]. Their bounded-domain discontinuous-Galerkin estimates are not an oracle for the present moving attractive Coulomb wells. We construct the needed conforming space and then certify its matrix arithmetic separately. More specifically, the products of tensor-linear hats with local polynomials below form a partition-of-unity approximation space, as in Melenk and Babuška [7, Section 2]. We give the cutoff and derivative estimates explicitly because the tensor cells can have arbitrarily large aspect ratios.

This is a form-domain approximation: the piecewise polynomial functions need only belong to H^1 . Their kinetic matrix uses first weak derivatives, not second derivatives across element boundaries.

Lemma 3.5 (Exponentially accurate Ritz approximation). *For each i , each rational choice of centers and secondary parameters, and each sufficiently large D , one can explicitly construct a space $\mathcal{V}_i \subset H^1(\mathbb{R}^3)$ of rational piecewise polynomials, of dimension polynomial in D , supported in $r_i < .8D$, such that*

$$\inf_{v \in \mathcal{V}_i} \|v - \psi_i\|_{H^1} \leq e^{-.58D}. \quad (3.25)$$

Its lowest Ritz value λ_i^R satisfies

$$0 \leq \lambda_i^R - \lambda_i \leq Ce^{-1.16D}. \quad (3.26)$$

The construction and constants are uniform over rational parameters throughout $[1, 4]^m$.

Proof. Work in local coordinates $z = y - X_i$ on $[-D, D]^3$, with poles at 0 and $\rho_i e_z$. Put $\delta = 2^{-8D}$ and fix a sufficiently small rational $\xi > 0$, as specified below. On each coordinate axis let B be the set of projections of the two poles. Starting from $[-D, D]$, bisect dyadically until every interval I obeys

$$|I| \leq \xi \min\{1, \text{dist}(I, B) + \delta\}. \quad (3.27)$$

Every leaf length is at least $\xi\delta/2$, since its parent failed the stopping rule. Above a fixed spacing there are $O(D)$ intervals. At a finer level of length h , only intervals within h/ξ of one of the at most two projections can still split, so a fixed number of intervals per level are split there. There are $O(\log(D/\delta)) = O(D + \log D)$ levels, hence $O(D)$ intervals per axis. This remains true for poles arbitrarily close to dyadic boundaries. The tensor product has $O(D^3)$ nodes and cells.

Let b_w be the tensor-linear hat at node w . The full hats form a partition of unity in the cube. Retain nodes satisfying

$$|w| < .73D, \quad |w| > \delta, \quad |w - \rho_i e_z| > \delta, \quad (3.28)$$

and take $b_w(z)(z - w)^\alpha$ for all $|\alpha| \leq L_0$, where L_0 is a sufficiently large fixed multiple of D . These functions span $\mathcal{V}_i$, whose dimension before removing dependencies is $O(D^6)$.

Here is the support estimate that avoids any aspect-ratio assumption. For either pole a , every coordinate interval adjacent to w obeys

$$|I| \leq \xi \min\{1, |w_j - a_j| + \delta\}.$$

The support radius R_w^{hat} of a retained hat thus satisfies

$$\begin{aligned} R_w^{\text{hat}} &\leq \xi(|w - a| + \sqrt{3}\delta) \leq (1 + \sqrt{3})\xi|w - a|, \\ R_w^{\text{hat}} &\leq \sqrt{3}\xi. \end{aligned} \quad (3.29)$$

For $d_w = \min\{1, |w|, |w - \rho_i e_z|\}$ this gives $R_w^{\text{hat}} \leq C\xi d_w$. Each support avoids both poles and, for large D , lies in $|z| < .8D$.

Choose a fixed $c < 1/10$ and set $R = cd_w$. The whole ball $B(w, R)$ avoids both poles. Throughout it every remote primary has distance at least $D - 2S - .73D - c > D/8$, so it also avoids every remote cap transition. On this enlarged ball the eigen-equation contains only pure Coulomb potentials. Rescaling by R gives $\Delta u = Fu$ on the unit ball. Its coefficient satisfies (3.20)

with an absolute A : local inverse-distance expansions give factorial derivative bounds because $R/\text{dist}(w, a) \leq c$; the extra R^2 in the coefficient makes $R^2/\text{dist}(w, a)$ bounded; and remote contributions sum to at most Cm/D . The eigenvalue term is bounded too. The sup norm of u is uniform by [Theorem 3.1](#).

Choose ξ sufficiently small compared to c and the fixed analytic radius in [Theorem 3.4](#), putting the hat supports inside that smaller ball after rescaling. With L_0 a sufficiently large fixed multiple of D , its Taylor polynomial p_w satisfies

$$|\psi_i - p_w| + R|\nabla\psi_i - \nabla p_w| \leq e^{-20D} \quad \text{on } \text{supp } b_w. \quad (3.30)$$

Increasing the fixed multiple absorbs the polynomial factor in the differentiated remainder. The Taylor coefficients establish existence of an approximant; the algorithm will not compute them.

To remove tiny neighborhoods of the poles and the exterior, let a_{out} equal one up to radius $.66D$, decrease linearly to zero by $.72D$, and vanish thereafter. Let a_{in} vanish up to δ , increase linearly to one by 2δ , and equal one thereafter. Sample

$$a_{\text{out}}(|z|) a_{\text{in}}(|z|) a_{\text{in}}(|z - \rho_i e_z|)$$

at all grid nodes to get weights $c_w \in [0, 1]$, and put $b = \sum_w c_w b_w$. Nonzero weights occur only at retained nodes. The function b equals one away from $O(\delta)$ -balls about the poles and away from $|z| > .65D$: a node within 2δ of a pole has adjacent lengths $O(\xi\delta)$, whereas all intervals have length at most ξ . These facts localize the inner and outer transitions, respectively.

Each first derivative of a tensor-linear interpolant is a convex combination of edge secants in that coordinate. Consequently $|\nabla b| \leq C/\delta$ on the inner transitions and $|\nabla b| \leq C/D$ on the outer transition, irrespective of aspect ratios. The bounded sup norm of ψ_i makes the inner cutoff-gradient cost $C\delta^{1/2}$. Also, by its H^2 bound,

$$\|\nabla\psi_i\|_{L^2(B_{C\delta}(a))} \leq C\delta\|\nabla\psi_i\|_6 \leq C\delta.$$

The exterior cost follows from [\(3.19\)](#). Thus

$$\|\psi_i - b\psi_i\|_{H^1} \leq C(\sqrt{\delta} + e^{-.61D}). \quad (3.31)$$

The cutoff weights, like the Taylor coefficients, are only existence coefficients in the explicitly generated space.

Set $v = \sum_w c_w b_w p_w \in \mathcal{V}_i$. At almost every point at most eight hats are nonzero. Their derivatives cost at most C/δ , by the minimum interval length, and $R^{-1} \leq C/\delta$. On a region of volume $O(D^3)$, [\(3.30\)](#) gives

$$\|b\psi_i - v\|_{H^1} \leq CD^{3/2}\delta^{-1}e^{-20D}.$$

Since $\sqrt{\delta} = e^{-4(\log 2)D}$ and $\delta^{-1}e^{-20D} = e^{-(20-8\log 2)D}$, this and [\(3.31\)](#) prove [\(3.25\)](#) for large D .

Finally, the form of $h_i - \lambda_i$ is bounded in absolute value by a constant times the H^1 norm squared. For $e = v - \psi_i$, the eigen-equation cancels the cross terms, giving

$$\frac{\langle v, h_i v \rangle}{\|v\|_2^2} - \lambda_i = \frac{\langle e, (h_i - \lambda_i)e \rangle}{\|v\|_2^2} \leq C\|e\|_{H^1}^2.$$

Here $\|v\|_2$ is bounded away from zero. The variational principle supplies the opposite inequality for the lowest Ritz value, proving [\(3.26\)](#). $\square$

3.6 Certified polynomial-bit eigenvalue evaluation

A small Ritz space by itself does not establish bit complexity. We next bound the conditioning loss and give certified matrix entries and eigenvalue enclosures. All arithmetic below is classical.

Proposition 3.6 (Uniform single-well eigenvalue oracle). *Suppose the centers and the parameters in $[1, 4]^m$ are rational, and all their numerators and denominators, as well as those of q, q' , have at most B bits. Under the hypotheses of this section, a deterministic algorithm returns rational numbers $\ell_i \leq \lambda_i \leq u_i$ with*

$$u_i - \ell_i \leq \varepsilon_D, \quad \varepsilon_D = 2^{-\lceil 8D/5 \rceil} < e^{-1.1D}. \quad (3.32)$$

Its bit complexity is polynomial in D, m, B . The bound is uniform over every rational sample in $[1, 4]^m$; no spectral gap of a Ritz matrix is assumed.

Proof. We divide the construction into exact polynomial arithmetic, certified Coulomb integration, and rational eigenvalue enclosure.

Exact matrices and a mass lower bound. Construct the basis in [Theorem 3.5](#). All grid decisions are rational comparisons: one-dimensional distances in [\(3.27\)](#) are rational, and node membership in [\(3.28\)](#) is checked by comparing squared distances. Grid depth is $O(D + \log D)$. Endpoints, hat coefficients, and coefficients of the degree- $O(D)$ monomials have polynomial bit length. For instance, raising a rational coordinate of b bits to degree L costs at most $O(Lb)$ bits, before polynomial sums; here $b = O(B + D + \log D)$ and $L = O(D)$. Only polynomially many coefficients and rational operations occur in any polynomial expansion or box integral. Summing rational terms by a product of their denominators gives a conservative polynomial bit bound even without exploiting common denominators.

Let $f_1, \dots, f_{r_0}$ be this possibly redundant basis, $r_0 = O(D^6)$. The mass and kinetic matrices

$$M_{ab} = \int f_a f_b \, dy, \quad T_{ab} = \frac{1}{2} \int \nabla f_a \cdot \nabla f_b \, dy \quad (3.33)$$

are exact rationals obtained by box integration. A product of two hats has only a fixed number of box pieces; alternatively, summing over the entire polynomial-size grid also suffices. Remove redundant functions by exact rational Gram-matrix elimination, retaining an independent subset. This gives a positive definite matrix M of dimension $r \leq r_0$, with corresponding kinetic matrix T . The subset is nonempty by the approximation property.

Let B_0 be the largest bit length of a numerator or positive denominator of the entries of this reduced M , enlarged to at least one. It is computed from the exact entries and is polynomial in D, m, B . If Q is the product of all entry denominators, then $Q \leq 2^{r^2 B_0}$ and QM is an integer positive definite matrix. Therefore $\det(QM) \geq 1$, while $\lambda_{\max}(M) \leq r 2^{B_0}$. The explicitly computable number

$$\mu = 2^{-K_M}, \quad K_M = r^3 B_0 + (r - 1)(B_0 + \lceil \log_2 r \rceil) \quad (3.34)$$

satisfies $\lambda_{\min}(M) \geq \mu$. This intentionally coarse determinant bound is enough: K_M is polynomial. Exact rank selection is itself polynomial-bit arithmetic. After clearing denominators, fraction-free elimination has entries given by minors of an integer matrix. Hadamard's determinant bound bounds their bit lengths by a polynomial in its dimension and entry bit length. Equivalently, ordinary rational elimination with fractions reduced at each step has the same polynomial bound through the bordered-minor formulas.

Certified Coulomb entries. The preceding determinant bound shows how many extra bits compensate for a poorly conditioned basis. It remains to compute each potential entry to that absolute accuracy. The graded supports avoid every pole, which permits a uniformly convergent kernel expansion on each box. On the supports of the basis functions every other cap equals one, so every

potential entry is a finite sum of pure Coulomb integrals against polynomials on rational boxes. For each such box Q_0 , each pole a , and its midpoint c_0 , put

$$s = |c_0 - a|^2, \quad z(y) = \frac{2(c_0 - a) \cdot (y - c_0) + |y - c_0|^2}{s}.$$

Every contributing box is in the support of a retained hat. By (3.29), its diameter is a fixed small fraction of its distance from either own pole. Remote poles are farther still. Decreasing the fixed ξ if necessary gives $|z(y)| \leq 1/2$ throughout every such box. Thus

$$\frac{1}{|y - a|} = s^{-1/2} \sum_{j=0}^J \binom{-1/2}{j} z(y)^j + \mathcal{R}_J(y), \quad |\mathcal{R}_J(y)| \leq s^{-1/2} 2^{-J}. \quad (3.35)$$

The coefficient bound $\left| \binom{-1/2}{j} \right| = 4^{-j} \binom{2j}{j} \leq 1$ proves this geometric remainder bound by summing the tail.

Here s and all coefficients of z are exact rationals. Both $\log_2^+(s^{-1/2})$ and the bit lengths of these rationals are polynomial in D, m, B ; indeed the own-pole distances on supports are bounded below by a fixed multiple of δ . The positive scalar $s^{-1/2}$ can be approximated to any requested polynomial number of bits by rational bisection and exact squaring.

We spell out an absolute-error certificate, so cancellation among polynomial terms causes no difficulty. If P is the polynomial multiplying a kernel on Q_0 , a rational coefficient-and-volume bound gives

$$A_{Q_0} \geq \int_{Q_0} |P(y)| dy.$$

One may take the volume times the sum of absolute coefficients, each multiplied by the appropriate coordinate bound to its degree. Its bit length is polynomial. Include the charge in this bound. If $s^{-1/2} \leq 2^b$, the truncation error is at most $A_{Q_0} 2^{b-J}$. An approximation to $s^{-1/2}$ of absolute error 2^{-q_0} adds at most $2A_{Q_0} 2^{-q_0}$, since the absolute sum in (3.35) is at most two. For T_0 box-pole summands, taking

$$\begin{aligned} J &\geq p + b + \lceil \log_2^+ A_{Q_0} \rceil + \lceil \log_2 T_0 \rceil + 3, \\ q_0 &\geq p + \lceil \log_2^+ A_{Q_0} \rceil + \lceil \log_2 T_0 \rceil + 4 \end{aligned}$$

makes their total error smaller than 2^{-p} . Use maxima over the summands to choose common orders. Here $\log_2^+ t = \max\{0, \log_2 t\}$; integer upper bounds can be read directly from rational numerator and denominator bit lengths.

The polynomial $z(y)^j$ has degree $2j$ in three variables. Even after multiplication by the basis polynomials it has only $O((D+J+1)^3)$ possible monomials. Polynomial multiplication, integration, and coefficient bit lengths are polynomial in D, m, B, J, q_0 . These bounds establish a deterministic entry algorithm with any prescribed polynomial number p of absolute precision bits. No matrix element is computed by integrating across a Coulomb singularity or a cap transition.

Matrix error and exact eigenvalue decisions. Let A be the exact form matrix of h_i in the independent basis. Compute a symmetric rational matrix $\tilde{A}$ by evaluating each potential entry, once per unordered index pair, to error at most

$$\tau = \frac{\varepsilon_D \mu}{16r}.$$

The kinetic matrix is exact. Since a symmetric matrix's operator norm is at most r times its largest absolute entry,

$$\|M^{-1/2}(\tilde{A} - A)M^{-1/2}\| \leq \varepsilon_D/16. \quad (3.36)$$

The required entry precision has $O(D + K_M + \log r)$ bits and is therefore polynomial. Let $\tilde{\lambda}$ be the minimum of the generalized Rayleigh quotient for $(\tilde{A}, M)$. The variational characterization and (3.36) give

$$|\tilde{\lambda} - \lambda_i^R| \leq \varepsilon_D/16.$$

An initial rational bracket is $[-R_0, R_0]$, where

$$R_0 = 1 + \frac{r \max_{a,b} |\tilde{A}_{ab}|}{\mu}.$$

Its bit length is polynomial. To bisect it at a rational z , test the inertia of $\tilde{A} - zM$ exactly. For clarity, an elementary exact inertia algorithm repeatedly uses congruence and Schur complements. A nonzero diagonal entry is a one-dimensional pivot and contributes its sign. If every remaining diagonal vanishes but an off-diagonal entry $a \neq 0$ remains, pivot on the block $\begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$, which contributes one positive and one negative direction. A zero remaining matrix contributes only zero directions. These operations give the full inertia. After denominators are cleared, each Schur entry is a ratio of bordered minors; determinant bounds again give polynomial bit growth. Thus this test is a polynomial-time exact rational calculation, including the case of zero eigenvalues.

Because M is positive definite, $\tilde{A} - zM$ is positive semidefinite exactly when $z \leq \tilde{\lambda}$. Update the lower or upper endpoint accordingly. Polynomially many bisections give a rational enclosure $[\ell, u]$ of $\tilde{\lambda}$ of width at most $\varepsilon_D/8$. Arbitrary proximity to a bisection point incurs no further precision requirement and no eigenvalue-gap assumption.

The fixed lower bound on D can be enlarged so that $Ce^{-1.16D} \leq \varepsilon_D/8$; this is possible since $(8/5) \log 2 < 1.16$. Combining the one-sided Ritz bias and the two-sided matrix error gives the certified interval

$$\lambda_i \in \left[\ell - \frac{\varepsilon_D}{16} - \frac{\varepsilon_D}{8}, u + \frac{\varepsilon_D}{16} \right]. \quad (3.37)$$

Its width is at most $3\varepsilon_D/8$. Notice that the Ritz bias is subtracted from the lower endpoint; a Ritz upper bound alone would not suffice for tuning. This proves the proposition.

All dimensions, operation counts, and working bit lengths are polynomial in D, m, B . The mesh and the independent subset may change at different rational samples, but every such sample obeys the same estimates. Constants controlling the polynomial degree, the cutoff size, and the minimum admissible D are fixed once from the uniform inequalities above. They are not input-dependent advice or a search for an unknown spectral gap. $\square$

3.7 Simultaneous rational tuning

The energy oracle now supplies certified intervals, while [Theorem 3.2](#) supplies the monotonicity and weak coupling of the tuning parameters. Combining these facts gives a rational algorithm: each coordinate is bracketed separately, and simultaneous updates contract. The stopping rule below uses an interval containing the target energy; it never asks for the exact sign of an unknown eigenvalue difference.

Proposition 3.7 (Tuning with fixed rounded charges). *At any fixed rational centers satisfying the hypotheses of this section, and with the integer charges in (3.2) already fixed, one can compute rational $\rho_1, \dots, \rho_m \in [1, 4]$ such that*

$$|\lambda_i + 1/2| \leq e^{-1.01D} \quad (1 \leq i \leq m). \quad (3.38)$$

The algorithm is deterministic and polynomial in D, m and the input bit lengths. Every parameter it samples or outputs has $O(D)$ fractional bits.

Proof. For frozen parameters other than ρ_i , the endpoint brackets, continuity, and strict secant bound in [Theorem 3.2](#) give a unique root of $\lambda_i = -1/2$ in $[1, 4]$. Let $T_i(\rho)$ be that root and let $T = (T_1, \dots, T_m)$, using the old other coordinates in every update. Combining [\(3.12\)](#) and [\(3.13\)](#) gives

$$\|T(\rho) - T(\tilde{\rho})\|_\infty \leq \theta \|\rho - \tilde{\rho}\|_\infty, \quad \theta = Cm/D^2 \leq 1/2. \quad (3.39)$$

Thus T has a unique fixed point ρ^* in the cube. This also follows directly by iterating the map: its successive differences are geometrically summable, and the limit is fixed by continuity.

We implement each coordinate update with a certified scalar bisection. Start with $[1, 4]$. At its midpoint invoke [Theorem 3.6](#). If the whole resulting eigenvalue interval lies below $-1/2$, retain the upper half of the parameter interval; if it lies above $-1/2$, retain the lower half. If it contains $-1/2$, stop and return that sample. In this last case the true energy differs from $-1/2$ by at most ε_D , so [\(3.12\)](#) puts the sample within $C\varepsilon_D/q'$ of the exact root. Otherwise, after $J = \lceil 4D \rceil$ bisections return the midpoint of the remaining parameter interval. All sign decisions were certified, so that interval still contains the root and has length $3 \cdot 2^{-J}$. Since $q' \leq 1$ and $2^{-4D} \ll \varepsilon_D$, the same update-error bound holds:

$$\|\hat{T}(\rho) - T(\rho)\|_\infty \leq C\varepsilon_D/q'. \quad (3.40)$$

This comparison-or-stop rule never requests an exact sign of an unknown irrational eigenvalue at a putative zero.

Start with all coordinates equal to two and perform J approximate simultaneous updates. If e_j is the sup-norm distance to ρ^* , then

$$e_{j+1} \leq \frac{1}{2}e_j + C\varepsilon_D/q', \quad e_J \leq 3 \cdot 2^{-J} + 2C\varepsilon_D/q'.$$

The own-coordinate upper slope and the other-coordinate variation bound together give

$$|\lambda_i(\rho) - \lambda_i(\rho^*)| \leq Cq' \|\rho - \rho^*\|_\infty.$$

Consequently the output has

$$|\lambda_i + 1/2| \leq Cq'2^{-J} + C\varepsilon_D \leq e^{-1.01D}$$

for large enough D , since $(8/5) \log 2 > 1.1 > 1.01$.

Each scalar bisection restarts from the rational interval $[1, 4]$ and makes at most $O(D)$ midpoint operations. Its samples, including early-return samples, therefore have $O(D)$ fractional bits independently of the number of simultaneous iterations. There are $O(mD^2)$ oracle calls. Their data bit lengths consist of the fixed-center and rounded-charge bit lengths together with these $O(D)$ parameter bits, so [Theorem 3.6](#) proves the total polynomial-time bound. In particular the charge-rounding error is part of every evaluated operator and is compensated by tuning; it is not introduced as an uncontrolled perturbation afterward. $\square$

4 Relative tails and computable hopping calibration

The absolute orbital estimate of [Proposition 3.1](#) does not determine an overlap of order e^{-D} . This section proves a relative tail estimate, identifies positive integrals that determine the hoppings, and shows how to calibrate those integrals before tuning the single-well energies. All constants below are independent of the site layout, D , the displacements, and the parameters $\rho_i \in [1, 4]$. The cap profile χ is fixed throughout.

We use the positions and operators from Section 3. In particular,

$$X_i = Dp_i + s_i, \quad \max_i |s_i| \leq S, \quad |X_i - X_j| \geq D - 2S \quad (i \neq j).$$

We take D sufficiently large and, for the estimates below, may assume $m + S \leq D^{1/4}$. Where a relative error is expanded, its displayed bound is assumed smaller than a fixed positive constant. The final parameter choice in Section 6 satisfies these conditions and makes every such error as small as required.

4.1 Relative comparison of single-well tails

For $r_i = |y - X_i|$ define the ray primitive

$$F_i(y) = r_i \int_0^1 W_i(X_i + t(y - X_i)) dt, \quad F_i(X_i) = 0. \quad (4.1)$$

The integral follows the entire segment from X_i to y ; it is not a local approximation to the field at either endpoint. Write

$$\varepsilon_{\text{rel}} = \frac{m \log D + m^2}{D}. \quad (4.2)$$

Proposition 4.1 (Uniform relative tail). *Suppose the actual rounded charges and the parameters ρ_i obey the conclusions of Proposition 3.7, in particular $|\lambda_i + 1/2| \leq e^{-1.01D}$. Uniformly for $r_i \leq 2D$,*

$$\psi_i(y) = \pi^{-1/2} e^{-r_i + F_i(y)} (1 + O(\varepsilon_{\text{rel}})). \quad (4.3)$$

Proof. Fix a site, suppress its subscript, and write $r = |y - X|$ and $\omega = (y - X)/r$ when $r > 0$. The capped-field estimates in Proposition 3.1 give

$$\|W\|_\infty \leq \frac{Cm}{D}, \quad \|\nabla W\|_\infty \leq \frac{Cm}{D^2}, \quad \|D^2 W\|_\infty \leq \frac{Cm}{D^3}. \quad (4.4)$$

Set $A(y) = \int_0^1 W(X + t(y - X)) dt$, so $F = rA$. For $0 < r \leq 3D$, differentiation gives

$$\begin{aligned} \nabla F &= \omega A + r \int_0^1 t \nabla W(X + t(y - X)) dt, \\ \Delta F &= \frac{2A}{r} + 2\omega \cdot \int_0^1 t \nabla W(X + t(y - X)) dt \\ &\quad + r \int_0^1 t^2 \Delta W(X + t(y - X)) dt. \end{aligned}$$

Consequently,

$$|\nabla F| \leq \frac{Cm}{D}, \quad |\Delta F| \leq C \left(\frac{m}{Dr} + \frac{m}{D^2} \right). \quad (4.5)$$

Also $F(r, \omega) = \int_0^r W(X + t\omega) dt$, and hence $\partial_r F = W(y)$ exactly.

Let $f = \pi^{-1/2} e^{-r+F}$. A direct calculation on the annulus $10 \leq r \leq 3D$ gives

$$\frac{(h_i - \lambda_i)f}{f} = -(\lambda_i + 1/2) + \frac{1}{r} - V_i - \frac{1}{2}(\Delta F + |\nabla F|^2), \quad (4.6)$$

$$\left| \frac{(h_i - \lambda_i)f}{f} \right| \leq C \left(\frac{m}{Dr} + \frac{m^2}{D^2} \right). \quad (4.7)$$

In the first equality the term $\partial_r F$ cancels W_i . For the second, note that $\rho_i \leq 4$ and $r \geq 10$, so

$$\left| \frac{1}{r} - V_i(y) \right| \leq \frac{|1-q|}{r} + \frac{q'}{r-4} \leq \frac{Cm}{Dr}.$$

The charge-rounding errors and $e^{-1.01D}$ are absorbed by the right-hand side of (4.7).

Choose a sufficiently large fixed constant A_0 and put

$$g(r) = A_0 \left[\frac{m}{D} (1 + \log(r/10)) + \frac{m^2 r}{D^2} \right]. \quad (4.8)$$

Then

$$g' = A_0 \left(\frac{m}{Dr} + \frac{m^2}{D^2} \right), \quad \Delta g = A_0 \left(\frac{m}{Dr^2} + \frac{2m^2}{D^2 r} \right), \quad 0 \leq \Delta g \leq \frac{2g'}{r}.$$

Replacing f by $fe^{\pm g}$ adds to its operator quotient the term

$$\pm(1 - \partial_r F)g' \mp \frac{1}{2}\Delta g - \frac{1}{2}(g')^2. \quad (4.9)$$

Since $r \geq 10$, $\partial_r F = O(m/D)$, and g' is uniformly small for sufficiently large D , the added term is at least cg' for the plus sign and at most $-cg'$ for the minus sign. Increasing A_0 dominates (4.7). Thus fe^g is a supersolution and fe^{-g} a subsolution.

At $r = 10$, Proposition 3.1 and Sobolev embedding give $\psi_i = \pi^{-1/2}e^{-10} + O(m/D)$, while $F = O(m/D)$. The reference value e^{-10} is a fixed positive number. A further fixed increase of A_0 therefore ensures $fe^{-g} \leq \psi_i \leq fe^g$ on the inner boundary. On the outer boundary $r = 3D$, Lemma 3.3 gives

$$|\psi_i| \leq Ce^{-0.99(3-0.01)D} = Ce^{-2.9601D}.$$

Moreover $0 \leq F \leq Cm$ there. With $\kappa = e^{-2.5D}$, the functions

$$f_+ = fe^g + \kappa, \quad f_- = fe^{-g} - \kappa$$

satisfy $f_- \leq \psi_i \leq f_+$ on both boundary spheres for large D . On the annulus the zero-order coefficient obeys

$$-V_i - W_i - \lambda_i \geq \frac{1}{2} - \frac{1}{10} - O(m/D) \geq \frac{1}{4}.$$

Adding the positive constant κ preserves the supersolution inequality, and subtracting it preserves the subsolution inequality. The maximum principle therefore yields the same inequalities throughout the annulus. For completeness, its weak form here follows by testing a supersolution-minus-solution with its nonnegative negative part: the resulting identity is the negative of a sum of a gradient norm and a strictly positive weighted L^2 norm, so the negative part vanishes. The same argument applies to a solution-minus-subsolution.

On $10 \leq r \leq 2D$, one has $g(r) = O(\varepsilon_{\text{rel}})$ and, since $F \geq 0$,

$$\frac{\kappa}{f} \leq \sqrt{\pi} e^{-D/2}.$$

The comparison inequalities imply (4.3) on this region. On $r \leq 10$, the reference exponential is bounded below by a fixed positive constant, $F = O(m/D)$, and the uniform absolute orbital estimate gives the same relative conclusion directly. $\square$

4.2 Surrogate fields and positive hopping integrals

We first identify the attraction integral to be calibrated. Write $V_k^{\text{hole}} = (1 - \chi_k)V_k$ for each site k , and for an ordered pair of distinct sites (i, j) define

$$B_{ij} = \langle \psi_j, V_j^{\text{hole}} \psi_i \rangle. \quad (4.10)$$

The identity

$$h = h_i - \sum_{k \neq i} V_k^{\text{hole}}$$

shows that the hole at j contributes $-B_{ij}$ to the matrix element $\langle \psi_j, (h + 1/2)\psi_i \rangle$ in the nonorthogonal orbitals. Section 5 will bound its remaining terms and the change under orthonormalization.

The relative tail formula still contains the secondary parameters, which have not yet been computed when the primary centers must be placed. We replace its capped field by an explicit one depending only on the primary centers. The replacement error is relative and uniform in the entire tuning cube; this is what makes the construction order possible.

Define

$$W_i^*(y) = \sum_{k \neq i} \frac{\chi(|y - X_k|/D)}{|y - X_k|}, \quad F_i^*(y) = r_i \int_0^1 W_i^*(X_i + t(y - X_i)) dt. \quad (4.11)$$

Each summand is extended by zero through its cutoff hole. In particular the apparent quotient at $y = X_k$ is identically zero in a neighborhood of that point.

Lemma 4.2 (Uniform capped kernels and surrogate replacement). *The zero-extended kernels*

$$G_D(z) = \frac{\chi(|z|/D)}{|z|}, \quad J_{D,\rho}(z) = \frac{\chi(|z|/D)}{|z - \rho e_z|}$$

are globally C^2 , uniformly for $1 \leq \rho \leq 4$. For every multi-index α with $|\alpha| \leq 2$,

$$\|\partial^\alpha G_D\|_\infty + \|\partial^\alpha J_{D,\rho}\|_\infty \leq CD^{-1-|\alpha|}. \quad (4.12)$$

Uniformly over all allowed secondary positions and rounded charges,

$$\|W_i - W_i^*\|_\infty \leq \frac{Cm^2}{D^2}, \quad |F_i(y) - F_i^*(y)| \leq \frac{Cm^2}{D} \quad (r_i \leq 3D). \quad (4.13)$$

Consequently, for tuned wells the relative formula (4.3) holds with F_i^* in place of F_i and the same error bound $O(\varepsilon_{\text{rel}})$.

Proof. For large D , both poles are strictly inside $|z| < D/16$, where the kernels vanish. On the nonzero part of either kernel, $|z| \geq D/16$ and $|z - \rho e_z| \geq |z|/2$. Rescale by $z = D\zeta$. Both kernels are D^{-1} times a function whose derivatives through order two are uniformly bounded: the shifted pole in the second rescaled kernel has position $(\rho/D)e_z$, a fixed positive distance from the support of χ . The C^2 matching of χ at its transition endpoints makes these global bounds, including at the hole boundaries. This proves (4.12).

Since $|q - 1| + q' \leq Cm/D$ after rounding,

$$W_i - W_i^* = \sum_{k \neq i} [(q - 1)G_D(y - X_k) + q'J_{D,\rho_k}(y - X_k)]$$

has sup norm $O(m^2/D^2)$. Integration over a ray of length at most $3D$ proves the second bound in (4.13). Exponentiating this bound in Proposition 4.1 proves the final assertion. None of the kernel estimates require tuning; they hold uniformly on the whole parameter cube. $\square$

For an oriented contact (i, j) , replacing the two orbital factors in B_{ij} by their relative approximations and the local nuclear potential by the unit primary kernel leads to the explicit positive integral

$$T_{ij}(X) = \frac{1}{\pi} \int_{|v| \leq D/8} \frac{1 - \chi(|v|/D)}{|v|} e^{L_X(v)} d^3v, \quad (4.14)$$

$$L_X(v) = -|X_j + v - X_i| - |v| + F_i^*(X_j + v) + F_j^*(X_j + v). \quad (4.15)$$

The fields in this definition use all centers in X . The integrable singularity $|v|^{-1}$ is part of the nonnegative integration weight, not part of the exponential. The next lemma justifies these replacements.

Lemma 4.3 (Comparison with the uncapped attraction). *For tuned wells and every contact,*

$$B_{ij} = T_{ij}(X)(1 + O(\varepsilon_{\text{rel}})). \quad (4.16)$$

The constants are uniform in $\rho_j \in [1, 4]$.

Proof. On the support of V_j^{hole} , write $y = X_j + v$. Then $|v| \leq D/8$ and $|y - X_i| \leq D + 2S + D/8 < 2D$. Proposition 4.1 and Lemma 4.2 apply to both wave functions. Their product is $\pi^{-1} e^{L_X(v)}$ times a uniform $1 + O(\varepsilon_{\text{rel}})$ factor. The primary kernel has coefficient $q = 1 + O(m/D)$, so it remains to compare the secondary kernel with the primary one in the resulting positive integral.

For $|v| > 10$,

$$\frac{|v|}{|v - \rho_j e_z|} \leq \frac{|v|}{|v| - 4} \leq \frac{5}{3}.$$

For $|v| \leq 10$, the hole weight is one when D is large. The ray derivative estimates, also valid for F^* , show that L_X is Lipschitz on this ball with a fixed constant. This statement requires no differentiability of $|v|$ at zero; it follows either by integrating the bounded derivatives away from zero or directly from the ray formula. Thus $e^{L_X(v)}$ is within fixed factors of $e^{L_X(0)}$ throughout the ball. Moreover,

$$\int_{|v| \leq 10} \frac{d^3v}{|v - \rho_j e_z|} \leq \int_{|w| \leq 14} \frac{d^3w}{|w|} < \infty, \quad \int_{|v| \leq 1} \frac{d^3v}{|v|} > 0.$$

The secondary-kernel integral on the inner ball is therefore at most a fixed multiple of the primary-kernel integral. Combining the two regions and multiplying by $q' = O(m/D)$ gives a relative $O(m/D)$ correction. Positivity allows the uniform relative wave-function errors to be taken outside both integrals. These errors are included in (4.16). $\square$

Lemma 4.4 (Size of the reference integrals). *Let $X^0 = (Dp_i)_i$. For every contact,*

$$ce^{-D} \leq T_{ij}(X^0) \leq CD^2 e^{-D+Cm}. \quad (4.17)$$

Proof. Put $u = u_{ij} = p_j - p_i$, so $|u| = 1$. The triangle inequality gives $|Du + v| + |v| \geq D$. Each ray correction in the integral is nonnegative and at most Cm , by (4.12) and its ray length. The integral of $|v|^{-1}$ over the ball of radius $D/8$ is $O(D^2)$, proving the upper bound. On $|v| \leq 1$ the hole weight is one, $|Du + v| + |v| \leq D + 2$, and the corrections are nonnegative. Integration over this fixed ball gives the lower bound; for example $2e^{-D-2}$ is a valid lower bound for large D . $\square$

4.3 Sensitivity to displacement

Lemma 4.5 (Uniform displacement law). *Let $X_i = Dp_i + s_i$ with $|s_i| \leq S$. There is a constant $c_1 > 0$ such that, uniformly on all contacts,*

$$\frac{T_{ij}(X)}{T_{ij}(X^0)} = e^{-u_{ij} \cdot (s_j - s_i)} (1 + O(\varepsilon_{\text{disp}})), \quad (4.18)$$

$$\varepsilon_{\text{disp}} = \frac{mS}{D} + \frac{S^2}{D} + SD^{-1/4} + e^{-c_1 \sqrt{D}}. \quad (4.19)$$

In particular the coefficient of the projected displacement in the leading exponent is one.

Proof. Compare the two integrals at the same v . In the ray defining $F_i^*(X_j + v)$, the reference path point $Dp_i + t(Dp_j + v - Dp_i)$ moves by $(1-t)s_i + ts_j$, of norm at most S . Each field center moves by at most S as well. The globally Lipschitz bound in (4.12) therefore changes the field along the path by at most CmS/D^2 . The ray has length $O(D)$ and its length changes by at most $2S$. Since the field itself is $O(m/D)$,

$$|F_i^*(X_j + v; X) - F_i^*(Dp_j + v; X^0)| \leq \frac{CmS}{D}. \quad (4.20)$$

The same bound holds for the F_j^* term; its ray length $|v|$ does not change. The global nature of the capped-kernel bound is essential here: it also applies when a moving ray crosses a hole or a transition interface.

Write $u = u_{ij}$, $\delta = s_j - s_i$, and $v = zu + w$ with $w \perp u$. Since $|v| \leq D/8$, $|Du + v| \geq 7D/8$. The Hessian of the norm on the segment from $Du + v$ to $Du + v + \delta$ has norm $O(1/D)$. Its gradient at $Du + v$ differs from u by $O(|w|/D)$. Taylor's formula gives

$$|Du + v + \delta| - |Du + v| = u \cdot \delta + O\left(\frac{S|w|}{D} + \frac{S^2}{D}\right). \quad (4.21)$$

This estimate includes the whole backward segment $v = -tu$, $0 \leq t \leq D/8$: there $Du + v = (D-t)u$ and the gradient is exactly u . Thus the segment's non-negligible longitudinal extent creates neither a different coefficient nor an $O(S)$ transverse error.

On the region $|w| \leq D^{3/4}$, Equations (4.20) and (4.21) show that

$$L_X(v) - L_{X^0}(v) + u \cdot \delta = O\left(\frac{mS}{D} + \frac{S^2}{D} + SD^{-1/4}\right). \quad (4.22)$$

The nonnegative integration weight is identical in the two integrals, so exponentiation controls their contributions on this region by the same relative factor.

On the remaining region $|w| > D^{3/4}$, one has

$$\begin{aligned} |Du + v| + |v| - D &\geq z + |v| = \frac{|w|^2}{|v| - z} \\ &\geq \frac{4|w|^2}{D} \geq 4\sqrt{D}. \end{aligned} \quad (4.23)$$

The identity is used only for $w \neq 0$, and the denominator is at most $2|v| \leq D/4$. The field corrections are bounded by Cm , without any angular constancy assumption. Displacement

changes a distance by at most $2S$, and $|u \cdot \delta| \leq 2S$. It follows that the omitted reference and displaced contributions, after division by e^{-D} and $e^{-D-u \cdot \delta}$ respectively, are at most

$$CD^2 \exp[-c\sqrt{D} + C(m + S)]. \quad (4.24)$$

By (4.17), these are also bounds on their contributions relative to the reference integral, with the appropriate displacement factor. The assumption $m + S \leq D^{1/4}$ and sufficiently large D absorb both the possibly large anisotropic factor e^{Cm} and the polynomial volume factor into $e^{-c_1\sqrt{D}}$. Combining this with (4.22), and using smallness of its right-hand side, proves the lemma. $\square$

4.4 Deterministic relative quadrature

We have reduced hopping calibration to positive integrals whose size is exponentially small. Direct absolute quadrature at that scale would obscure the complexity bound. Instead we control the oscillation of the logarithm of the exponential factor, while integrating its nonnegative radial weight exactly. Positivity then converts local errors into a relative error for the whole integral.

Lemma 4.6 (Computing the positive reference integrals). *For $0 < \epsilon < 1/10$, one can compute a positive rational $\hat{I}_{ij}$ satisfying*

$$\left| \log \frac{\hat{I}_{ij}}{e^{DT_{ij}(X^0)}} \right| \leq \epsilon \quad (4.25)$$

by a deterministic algorithm polynomial in D , m , the bit lengths of the layout data, and ϵ^{-1} . In particular the algorithm is polynomial in the source size when D is a fixed polynomial and ϵ an inverse polynomial.

Proof. Let $L = L_{X^0}$ in (4.15). On the integration ball, the two distance terms have Lipschitz constants one, and the ray formulas and (4.12) give a Lipschitz constant Cm/D for each correction. Hence

$$|L(v) - L(v')| \leq C|v - v'|, \quad |L(v)| \leq C(D + m). \quad (4.26)$$

Only the exponential factor has been included in this logarithm. No lower bound on $1 - \chi$ or differentiability of $\log(1 - \chi)$ is needed.

We describe a mesh with exact positive weights. Parameterize the unit sphere, up to a set of area zero, by

$$\omega(\mu, \varphi) = (\sqrt{1 - \mu^2} \cos \varphi, \sqrt{1 - \mu^2} \sin \varphi, \mu).$$

Here $-1 \leq \mu \leq 1$, $0 \leq \varphi \leq 2\pi$, and $d\omega = d\mu d\varphi$. Use M equal intervals in μ , N equal intervals in φ , and a rational radial partition with mesh at most h . Insert all the finitely many breakpoints of $\chi(r/D)$ into that partition. On a radial interval I_a , the weight

$$w_a = \int_{I_a} (1 - \chi(r/D)) r \, dr$$

is a nonnegative exact rational, computable by polynomial integration. Discard intervals with zero weight. The normalized angular weight of every angular rectangle is exactly

$$\frac{1}{\pi} \frac{2}{M} \frac{2\pi}{N} = \frac{4}{MN}. \quad (4.27)$$

Choose one sample v_{abc} in each resulting spatial cell, for example at the parameter midpoints.

The square-root parameterization satisfies

$$|\omega(\mu, \varphi) - \omega(\mu', \varphi')| \leq C|\mu - \mu'|^{1/2} + C|\varphi - \varphi'|.$$

Therefore $M = O((D/h)^2)$ and $N = O(D/h)$, with sufficiently large fixed constants, make every spatial cell have diameter $O(h)$, including cells adjacent to the poles. There are $O((D/h)^4)$ cells, a polynomial number. Their exact weighted sample sum is

$$Q = \sum_{a,b,c} \frac{4w_a}{MN} e^{L(v_{abc})}.$$

Positivity and (4.26) imply

$$e^{-Ch}Q \leq T_{ij}(X^0) \leq e^{Ch}Q. \quad (4.28)$$

Thus a fixed multiple of ϵ is sufficient for h . This relative estimate is independent of the size e^{-D} of the integral.

It remains to specify evaluation and bit accuracy. For a ray of length $r \leq 2D$, the integrand $t \mapsto W_i^*(X_i + t(y - X_i))$ has Lipschitz constant at most Cmr/D^2 . A midpoint rule with K equal panels therefore approximates its ray primitive with error at most

$$C \frac{mr^2}{D^2K} \leq \frac{Cm}{K}. \quad (4.29)$$

Taking $K = O(m/\epsilon)$ and evaluating each field sample to absolute error $O(\epsilon/D)$ gives an $O(\epsilon)$ error in each primitive. Each field sample is a sum of $m - 1$ globally Lipschitz capped kernels. At small radius the kernel is evaluated as zero, and elsewhere its denominator is bounded below by a fixed multiple of D ; equivalently one evaluates the fixed continuous function $\chi(a)/a$, extended by zero on $[0, 1/16]$, at $a = |z|/D$. Its global Lipschitz bound controls errors even if the approximate radius lies on the other side of a transition breakpoint. No crossing location needs to be found.

The coordinates Dp_i , sample coordinates, square roots, and trigonometric values can be computed to inverse-polynomial absolute accuracy with polynomially many bits. The global kernel and ray bounds just proved control the resulting errors. Thus all sample logarithms can be approximated to an arbitrarily specified $O(\epsilon)$ absolute error with polynomial work. Such logarithm errors change the positive sample sum by a factor $e^{O(\epsilon)}$, not by an additive error multiplied by an uncontrolled exponential.

For the remaining arithmetic, compute the scaled sum $e^D Q$, using sample exponentials $e^{D+L(v_{abc})}$. Its exact value is bounded below by a positive constant up to the factor in (4.28), because (4.17) gives $e^D T_{ij}(X^0) \geq 2e^{-2}$. The sum of its non-exponential weights is $O(D^2)$, and all exponent arguments have magnitude $O(D + m)$. After the logarithms have been approximated as above, each elementary exponential can be rounded to sufficiently small absolute error $O(\epsilon/D^2)$, clipping a negative rounded value to zero; this adds $O(\epsilon)$ to the scaled sum. These last approximations require only $O(D + m + \log(D/\epsilon))$ precision bits, with polynomial working precision. This step does not require exponentially accurate ray quadrature: the earlier ray errors have already been bounded multiplicatively. Standard range reduction and Taylor series compute the square roots, trigonometric values, exponentials, and logarithms to these precisions in polynomial time. The exact rational weights have polynomial bit lengths, and the sum contains polynomially many terms.

Allocate a sufficiently small fixed fraction of ϵ to each of the spatial, ray, coordinate, and arithmetic errors. The resulting positive rational sum is $\hat{I}_{ij}$ and satisfies (4.25). $\square$

4.5 Calibration, rounding, and orientation

The oriented attraction integrals are close to a symmetric matrix for an exact reason that is useful independently of their asymptotic formula.

Lemma 4.7 (Orientation identity). *For distinct sites,*

$$B_{ij} - B_{ji} = (\lambda_i - \lambda_j) \langle \psi_j, \psi_i \rangle. \quad (4.30)$$

For tuned wells and large D under the stated separation bounds,

$$|\langle \psi_j, \psi_i \rangle| \leq e^{-0.85D}, \quad |B_{ij} - B_{ji}| \leq 2e^{-1.86D}. \quad (4.31)$$

Proof. The single-well operators have common domain H^2 , and

$$h_i - h_j = V_j^{\text{hole}} - V_i^{\text{hole}}.$$

Pair this identity between the real eigenfunctions ψ_j and ψ_i , using self-adjointness of h_j , to obtain (4.30). All the pairings are well-defined by the Coulomb domain bounds of Proposition 3.1. Equivalently, in the two expansions of the full matrix element $\langle \psi_j, (h + 1/2)\psi_i \rangle$, contributions from every hole other than i and j cancel identically, because they integrate the same real product against the same multiplication potential.

To record an explicit overlap bound, Lemma 3.3 gives

$$|\psi_i(y)\psi_j(y)| \leq Ce^{0.0198D}e^{-0.99(r_i+r_j)}.$$

Use $r_i + r_j \geq D - 2S$ on a factor with exponent 0.90, and retain $e^{-0.09r_i}$ for integrability. It follows that

$$|\langle \psi_j, \psi_i \rangle| \leq Ce^{-0.8802D+1.8S} \int_{\mathbb{R}^3} e^{-0.09|y-X_i|} d^3y \leq e^{-0.85D}$$

for sufficiently large D . Finally $|\lambda_i - \lambda_j| \leq 2e^{-1.01D}$ proves the second bound. $\square$

For a tolerance $\tau > 0$, define the total calibration error

$$\begin{aligned} \mathcal{E}_{\text{cal}} &= \frac{m \log D + m^2}{D} + \frac{mS}{D} + \frac{S^2}{D} \\ &\quad + SD^{-1/4} + \tau + e^{-c_1\sqrt{D}}. \end{aligned} \quad (4.32)$$

Proposition 4.8 (Deterministic hopping calibration). *Let the contact layout be as in Lemma 2.4, choose one orientation per contact, and let the prescribed rational hoppings satisfy $0 < t_{\min} \leq t_{ij} \leq 1$, where $t_{\min}$ is an inverse polynomial in the source size. Let Z_* be an integer with $e^D/2 \leq Z_* \leq 2e^D$ and round the nuclear charges as in Section 3. For inverse-polynomial τ , there is a deterministic polynomial-time procedure that computes rational centers $X_i = Dp_i + s_i$ obeying*

$$S := \max_i |s_i| \leq Cm(m + \log D + \log(1/t_{\min}) + 1) + \tau. \quad (4.33)$$

After applying Proposition 3.7 at these final centers, the resulting orbitals satisfy, on every chosen oriented contact,

$$|Z_*B_{ij} - t_{ij}| \leq Ct_{ij}\mathcal{E}_{\text{cal}}. \quad (4.34)$$

Here D is taken sufficiently large for the hypotheses above and for $\mathcal{E}_{\text{cal}}$ to be small. The reverse oriented attraction has an additional error at most $4e^{-0.86D}$ in the same rescaled units. No additional geometric equation for the reverse orientation is needed.

Proof. Compute, to additive error at most $\tau/4$, the quantities

$$d_{ij} = \log \left(\frac{Z_* T_{ij}(X^0)}{t_{ij}} \right), \quad X^0 = (Dp_i)_i. \quad (4.35)$$

Lemma 4.6 supplies the requisite relative integral accuracy. One can evaluate the logarithm as

$$\log Z_* - D + \log(e^D T_{ij}(X^0)) - \log t_{ij},$$

so all elementary-function calculations have polynomial bit cost. The two-sided bound (4.17) yields

$$\max_{(i,j) \in \mathcal{C}} |d_{ij}| \leq C(m + \log D + \log(1/t_{\min}) + 1). \quad (4.36)$$

Apply the constructive linear right inverse in Lemma 2.5 to the approximated data. Its operator bound is Cm from the maximum edge datum to the maximum site displacement. Its formulas use rational operations and fixed elements of $\mathbb{Q}(\sqrt{3})$, so they can be evaluated with polynomially many bits. Round the resulting positions $Dp_i + s_i$ to rational positions with Euclidean site error at most $\tau/8$, allocating another such allowance if approximate arithmetic is used in the right inverse. Define the final displacements from these rational positions by $s_i = X_i - Dp_i$. This definition preserves the exact contact vector decomposition even though s_i need not be rational. Since every u_{ij} is a unit vector, the resulting equations satisfy

$$|u_{ij} \cdot (s_j - s_i) - d_{ij}| \leq \tau. \quad (4.37)$$

The right-inverse bound and (4.36) imply (4.33). For example, the $\sqrt{3}$ arithmetic may be done to absolute accuracy $O(\tau/[D + \text{poly}(m) \max(1, \max |d_{ij}|)])$; the required number of bits is polynomial. Noncontacts retain separation at least $\sqrt{2}D - 2S$ by the triangle inequality.

There is no requirement to realize prescribed distances exactly. For a contact with $\delta = s_j - s_i$ the actual distance is

$$|Du_{ij} + \delta| = D + u_{ij} \cdot \delta + O(S^2/D).$$

Actual coordinates automatically satisfy all vector-cycle identities; the displayed per-edge correction is already included in Lemma 4.5 and is not accumulated through a further nonlinear solve.

Put $e_{ij} = u_{ij} \cdot (s_j - s_i) - d_{ij}$. The key cancellation is the exact identity

$$Z_* T_{ij}(X^0) e^{-u_{ij} \cdot (s_j - s_i)} = t_{ij} e^{-e_{ij}}, \quad |e_{ij}| \leq \tau. \quad (4.38)$$

Lemma 4.5 therefore gives $Z_* T_{ij}(X) = t_{ij}(1 + O(\varepsilon_{\text{disp}} + \tau))$. After tuning, Lemma 4.3 adds only $O(\varepsilon_{\text{rel}})$ relative error. This proves (4.34). In particular no factor e^{Cm} or $e^{|d_{ij}|}$ multiplies its absolute error, even for the smallest floored hopping. Since $t_{ij} \leq 1$, the absolute error per contact is at most $C\mathcal{E}_{\text{cal}}$; summing hopping operators incurs only polynomial factors in the number of sites.

All computations of X used the surrogate fields W_i^* . They depend on the primary centers and on D , but not on any eigenfunction or on the secondary parameters ρ_i . Thus the algorithm first fixes the rounded charges, computes and rounds these centers, and only then invokes Proposition 3.7 at the final rational centers. The surrogate-field replacement bounds hold uniformly over the tuning cube, and the relative orbital comparisons hold uniformly over all tuned outputs. Thus tuning at these fixed centers requires no recalibration and introduces no subsequent position-rounding error.

Finally Lemma 4.7 and $Z_* \leq 2e^D$ give $Z_* |B_{ij} - B_{ji}| \leq 4e^{-0.86D}$. The common target energy $-1/2$ is sufficient; exact equality of the λ_i is not required. The full one-body operator is Hermitian, so one may use either orientation for its off-diagonal entry after the negligible other-hole terms are accounted for in Section 5. $\square$

If the contact set is empty, the procedure simply rounds Dp_i to rational centers at the stated site accuracy; there are no hopping equations to solve. In general (4.33) is $S \leq \text{poly}(n)(1 + \log D)$. The bounds needed for the global choice are precisely (4.32), together with $m + S \leq D^{1/4}$ and the exponentially small orientation error. All scale choices and the conversion to the final many-electron energy error are made in Section 6.

5 Excluding lower-energy continuum states

We now compare the spectral infimum of the full Coulomb operator with the half-filled Hubbard energy. In particular, the comparison below includes a lower bound on every state outside the selected orbital space.

The proof has three distinct outputs. We first identify the one-particle matrix in orthonormal site modes and prove a fixed positive gap on its entire orthogonal complement. We then bound the many-electron coupling to that complement and eliminate it by a quadratic-form square completion. Only after this all-state lower bound is available do we compare the compressed Coulomb interaction with the Hubbard onsite term. Keeping these steps separate prevents a variational upper bound from being used as a NO-instance lower bound.

We use the rational centers and tuned parameters constructed in Sections 3 and 4. Increasing the fixed polynomial choice of D if necessary, we may assume throughout this section that

$$m + S \leq D^{1/4}, \quad D \geq D_0,$$

where the constant D_0 can be chosen uniformly. The estimates from Propositions 3.1 and 3.7 and Lemma 3.3 that we will use are

$$\|\psi_i - \phi_i\|_{H^2} \leq Cm/D, \quad \|\psi_i\|_{H^2} \leq C, \quad (5.1)$$

$$|\lambda_i + 1/2| \leq e^{-1.01D}, \quad |\psi_i(y)| \leq Ce^{-.99(r_i - .01D)}. \quad (5.2)$$

There is also a constant $g_w > 0$, independent of the construction parameters, such that

$$h_i - \lambda_i \geq g_w(I - |\psi_i\rangle\langle\psi_i|). \quad (5.3)$$

All operator inequalities involving an unbounded operator are understood as quadratic-form inequalities. By Lemma 2.4 and the displacement bound,

$$|X_i - X_j| \geq D - 2S \quad (i \neq j), \quad |X_i - X_j| \geq \sqrt{2}D - 2S \quad (\{i, j\} \notin \mathcal{C}). \quad (5.4)$$

The second inequality concerns distinct noncontact sites.

5.1 The one-particle matrix

Recall the hole potentials from Section 4, and write

$$A = h + \frac{1}{2}, \quad V_i^{\text{hole}} = (1 - \chi_i)V_i, \quad \epsilon_i = \lambda_i + \frac{1}{2}.$$

For the chosen orientation $i \rightarrow j$ of a contact, define

$$B_{ij} = \langle \psi_j, V_j^{\text{hole}} \psi_i \rangle, \quad \epsilon_{\text{hop}} = \max_{i \rightarrow j} |Z_* B_{ij} - t_{ij}|.$$

The maximum is zero if there are no contacts. Proposition 4.8 gives

$$\epsilon_{\text{hop}} \leq C\mathcal{E}_{\text{cal}}, \quad (5.5)$$

because all the target hoppings are at most one. Let T be the real symmetric m -by- m matrix with entry $-t_{ij}$ on contacts and zero elsewhere, including its diagonal.

Define $\mathcal{U} : \mathbb{C}^m \rightarrow L^2(\mathbb{R}^3)$ by $\mathcal{U}e_i = \psi_i$, and let $G = \mathcal{U}^*\mathcal{U}$ be its Gram matrix. The notation $\mathcal{U}$ denotes this map; the scalar onsite repulsion remains $U = 5/8$.

Lemma 5.1 (Residuals and the orthonormalized matrix). *For the preceding parameters and sufficiently large D ,*

$$\|A\psi_i\|_2 \leq e^{-.75D}, \quad \|G - I\| \leq e^{-.85D}. \quad (5.6)$$

In particular G is positive definite. The isometry $\mathcal{J}_0 = \mathcal{U}G^{-1/2}$ has orthonormal columns $\omega_i = \mathcal{J}_0e_i$, spanning the same spatial orbital space. Its matrix satisfies

$$\|Z_*\mathcal{J}_0^*A\mathcal{J}_0 - \mathsf{T}\| \leq \delta_1, \quad (5.7)$$

$$\delta_1 := m\epsilon_{\text{hop}} + Z_*e^{-1.01D} + CmZ_*e^{-1.1D} + CmZ_*e^{-1.6D}.$$

If $P_0 = \mathcal{J}_0\mathcal{J}_0^*$, then

$$\|AP_0\| \leq C\sqrt{m}e^{-.75D}, \quad \|\omega_i - \phi_i\|_{H^1} \leq Cm/D. \quad (5.8)$$

Proof. Since $h = h_i - \sum_{k \neq i} V_k^{\text{hole}}$,

$$A\psi_i = \epsilon_i\psi_i - \sum_{k \neq i} V_k^{\text{hole}}\psi_i. \quad (5.9)$$

The k -th hole is supported in $r_k \leq D/8$. On it,

$$r_i \geq 7D/8 - 2S \quad (k \neq i).$$

Both Coulomb poles of V_k lie in this hole for large D . The charges q, q' are bounded, and direct radial integration, enlarging the integration ball by at most four for the secondary pole, gives

$$\|V_k^{\text{hole}}\|_2 \leq CD^{1/2}, \quad \|V_k^{\text{hole}}\|_1 \leq CD^2. \quad (5.10)$$

Consequently the sum in (5.9) has norm at most

$$CmD^{1/2} \exp(-.85635D + 1.98S).$$

Together with the tuning error, this is at most $e^{-.75D}$ for the stated regime and sufficiently large D .

For $i \neq j$, put $\ell_{ij} = |X_i - X_j|$. The product of the two tail bounds is at most

$$C \exp(-.99(r_i + r_j) + .0198D).$$

Use $r_i + r_j \geq \ell_{ij}$ on a factor with exponent .90; the remaining factor is integrable uniformly in the centers, since $\int e^{-.09(r_i+r_j)} dy \leq \int e^{-.09r_i} dy \leq C$. Thus

$$|G_{ij}| \leq Ce^{-.90\ell_{ij}+.0198D} \leq Ce^{-.8802D+1.8S}.$$

The diagonal entries are one. The row-sum estimate and $m + S \leq D^{1/4}$ prove the second inequality in (5.6).

Let $M = \mathcal{U}^* A \mathcal{U}$, which is real symmetric. We next estimate its entries before orthonormalization. If $k \neq i, j$, both r_i and r_j are at least $7D/8 - 2S$ on the k -th hole. Equations (5.2) and (5.10) therefore imply

$$\sum_{k \neq i, j} |\langle \psi_j, V_k^{\text{hole}} \psi_i \rangle| \leq C m D^2 e^{-1.7127D + 3.96S}.$$

The same estimate bounds $\sum_{k \neq i} \langle \psi_i, V_k^{\text{hole}} \psi_i \rangle$. For a noncontact pair, the remaining hole at j is bounded using $r_i + r_j \geq \sqrt{2}D - 2S$, giving

$$|\langle \psi_j, V_j^{\text{hole}} \psi_i \rangle| \leq C D^2 e^{-(.99\sqrt{2} - .0198)D + 1.98S}.$$

Here $.99\sqrt{2} - .0198 > 1.38$. All these bounds are smaller than $e^{-1.1D}$, after absorbing their polynomial factors and fixed constants. For $i \neq j$, the term $\epsilon_i G_{ji}$ is at most $e^{-1.86D}$. It follows, adjusting a fixed constant, that

$$\begin{aligned} |M_{ii} - \epsilon_i| &\leq C e^{-1.1D}, \\ |M_{ji} + B_{ij}| &\leq C e^{-1.1D} & (i \rightarrow j \text{ a chosen contact}), \\ |M_{ij}| &\leq C e^{-1.1D} & (i \neq j, \{i, j\} \notin \mathcal{C}). \end{aligned} \quad (5.11)$$

The exactly symmetric matrix M requires only one of the two orientations of each contact. Its diagonal tuning contribution has norm at most $e^{-1.01D}$, not $m e^{-1.01D}$. Entrywise comparison with $\mathbb{T}$ and a row-sum bound now give

$$\|Z_* M - \mathbb{T}\| \leq m \epsilon_{\text{hop}} + Z_* e^{-1.01D} + C m Z_* e^{-1.1D}.$$

For completeness, the residual estimate controls the entire matrix, without using a dimension-dependent bound on a many-body space:

$$\|A \mathcal{U}\| \leq \sqrt{m} e^{-.75D}, \quad \|\mathcal{U}\| = \|G\|^{1/2} \leq 2, \quad \|M\| \leq C \sqrt{m} e^{-.75D}.$$

Functional calculus on the positive matrix G gives $\|G^{-1/2} - I\| \leq C e^{-.85D}$ and $\|G^{-1/2}\| \leq 2$. Consequently

$$\|G^{-1/2} M G^{-1/2} - M\| \leq C \sqrt{m} e^{-1.6D} \leq C m e^{-1.6D}. \quad (5.12)$$

This orthonormalizes the matrix of the shifted operator $h + 1/2$. In the original nonorthogonal modes, $M = \mathcal{U}^* h \mathcal{U} + \frac{1}{2} G$. The shift is essential in this estimate: one must add $\frac{1}{2} G$, rather than $\frac{1}{2} I$, before orthonormalization. Equation (5.7) follows.

The same residual estimate applied to $A \mathcal{J}_0 = A \mathcal{U} G^{-1/2}$ proves the first part of (5.8). Finally, as a map into H^1 , $\mathcal{U}$ has norm at most $C \sqrt{m}$, by (5.1). Thus

$$\|\omega_i - \psi_i\|_{H^1} \leq C \sqrt{m} e^{-.85D}.$$

Combining this with $\|\psi_i - \phi_i\|_{H^1} \leq C m / D$ gives the remaining assertion. $\square$

We now include spin. Set

$$\mathfrak{h} = L^2(\mathbb{R}^3; \mathbb{C}^2), \quad P = P_0 \otimes I_2, \quad \mathcal{J} = \mathcal{J}_0 \otimes I_2.$$

Thus P_0 has rank m , whereas P has rank $2m$. The operator A acts trivially on spin; we use the same symbol for this extension. Its residual and matrix bounds are unchanged.

5.2 A gap on the entire one-particle complement

Spatial localization separates the nuclei from the exterior, where the attraction is uniformly weak. We use the IMS product-rule identity (compare Simon [14, Lemma 3.1]); its kinetic-error coefficient is $1/2$ for our normalization $-\Delta/2$. The identity is derived below. The additional point to verify is that a globally orthogonal function has only exponentially small components along the local ground states after localization.

Proposition 5.2 (Continuum complement gap). *There is a constant $g > 0$ such that, for all sufficiently large D ,*

$$(I - P)A(I - P) \geq g(I - P).$$

The inequality holds on the full one-particle form domain, including functions supported between the nuclei or at infinity.

Proof. Choose smooth real functions $\theta_0, \theta_1, \dots, \theta_m$ with

$$\sum_{a=0}^m \theta_a^2 = 1, \quad \theta_i = 1 \text{ on } r_i \leq D/5, \quad \text{supp } \theta_i \subseteq \{r_i \leq D/4\}.$$

The balls of radius $D/4$ are disjoint by (5.4). In the annulus $D/5 < r_i < D/4$, take $\theta_i = \cos f(r_i/D)$ and $\theta_0 = \sin f(r_i/D)$, where f is a fixed smooth transition from zero to $\pi/2$, constant near both endpoints. Set $\theta_0 = 0$ in the inner balls and one outside the outer balls. At most one transition annulus is present at any point, so

$$\sum_{a=0}^m |\nabla \theta_a|^2 \leq CD^{-2}. \quad (5.13)$$

This construction also verifies smoothness at the boundaries.

Let $u \in H^1(\mathbb{R}^3; \mathbb{C}^2)$, $Pu = 0$, and $\|u\| = 1$. In spin component σ , global orthogonality implies

$$b_{i\sigma} := \langle \psi_i, \theta_i u_\sigma \rangle = \langle (\theta_i - 1) \psi_i, u_\sigma \rangle.$$

The tail estimate, integrated over $r_i \geq D/5$, gives

$$\|(\theta_i - 1) \psi_i\| \leq C \text{poly}(D) e^{-.99(.2-.01)D} \leq e^{-.18D}.$$

Here the polynomial comes only from radial integration and has fixed degree. Consequently

$$\sum_{i,\sigma} |b_{i\sigma}|^2 \leq m e^{-.36D}. \quad (5.14)$$

There is no assumption that a localized vector is exactly orthogonal to the local ground state.

On $\text{supp } \theta_i$, every $k \neq i$ satisfies

$$r_k \geq D - 2S - D/4 = 3D/4 - 2S > D/8.$$

Hence $\chi_k = 1$ throughout that support and $h = h_i$ there. Choose a fixed $0 < g_{\text{loc}} \leq \min(g_w/2, 1/4)$. Equations (5.3) and (5.2), for large D , imply

$$\langle \theta_i u, A \theta_i u \rangle \geq g_{\text{loc}} \|\theta_i u\|^2 - (g_{\text{loc}} + e^{-1.01D}) \sum_{\sigma} |b_{i\sigma}|^2.$$

On $\text{supp } \theta_0$, each primary distance is at least $D/5$. Each secondary distance is at least $D/5 - \rho_i \geq D/5 - 4 \geq D/6$. Since $q + q' \leq 2$ for large D , the magnitude of the full attraction on this support is at most Cm/D . Thus

$$\langle \theta_0 u, A \theta_0 u \rangle \geq (1/2 - Cm/D) \|\theta_0 u\|^2 \geq g_{\text{loc}} \|\theta_0 u\|^2.$$

The kinetic product rule yields the IMS identity

$$\langle u, Au \rangle = \sum_{a=0}^m \langle \theta_a u, A \theta_a u \rangle - \frac{1}{2} \int \sum_{a=0}^m |\nabla \theta_a|^2 |u|^2.$$

It holds first for smooth functions and then on H^1 by continuity of the Coulomb forms. Summing the preceding bounds and using (5.13) and (5.14) gives

$$\langle u, Au \rangle \geq g_{\text{loc}} - Cme^{-.36D} - CD^{-2} \geq g_{\text{loc}}/2.$$

Taking $g = g_{\text{loc}}/2$ proves the proposition. $\square$

5.3 Uniform many-fermion bounds and elimination of high modes

Let

$$\mathfrak{H}_m = \bigwedge^m \mathfrak{h}, \quad \Pi = \bigwedge^m P, \quad Q_m = I - \Pi.$$

Here Π projects onto *all* m -electron configurations in the $2m$ spin orbitals; it does not impose single occupancy or a total-spin condition. Denote second quantization on this fixed particle-number space by $d\Gamma(B) = \sum_{\ell=1}^m B^{(\ell)}$. In the scaled coordinates, write $W_{\text{ee}} = \sum_{1 \leq \ell < k \leq m} |y_\ell - y_k|^{-1}$ for the electron repulsion, acting trivially on spin.

Under the unitary dilation $y_\ell = Z_* x_\ell$, the physical Hamiltonian obeys the exact relation

$$\frac{H}{Z_*^2} = d\Gamma(h) + \frac{W_{\text{ee}}}{Z_*}. \quad (5.15)$$

We henceforth use this unitary identification and set

$$\mathcal{H} = \frac{H}{Z_*^2} + \frac{m}{2} = d\Gamma(A) + \frac{W_{\text{ee}}}{Z_*}. \quad (5.16)$$

To compare its spectral infimum with its compression to Π , we will bound the gap on Q_m and the coupling $Q_m \mathcal{H} \Pi$. The one-particle estimates control the contribution from $d\Gamma(A)$; the following Hardy estimate controls the repulsion on every vector in $\text{Ran } \Pi$.

We briefly specify the domains used below. On the antisymmetric subspace, the kinetic form domain is H^1 in all $3m$ coordinates, with its finite spin components. Its operator domain is H^2 in those coordinates. Translated Hardy inequalities in one particle coordinate show that each nuclear or pair Coulomb multiplier is bounded from H^1 to L^2 . For a fixed instance their finite sum therefore satisfies $\|Vu\|_2 \leq C_{\text{inst}} \|u\|_{H^1}$. The Fourier inequality $\|\nabla u\|_2 \leq \varepsilon \|\Delta u\|_2 + C_\varepsilon \|u\|_2$ makes the potential infinitesimally operator-bounded relative to the total Laplacian. The Coulomb operator consequently has the indicated H^2 operator domain and H^1 form domain. The same reasoning applies to the one-particle operators. Every ω_i is in $H^2(\mathbb{R}^3)$. Products of these functions have square-integrable pure second derivatives and mixed first derivatives, so every vector in $\text{Ran } \Pi$ belongs to the many-particle operator domain. Both Π and Q_m preserve the form domain.

Lemma 5.3 (A dimension-independent fermionic Hardy bound). *Uniformly over all normalized $u \in \text{Ran } \Pi$,*

$$\sum_{\ell=1}^m \|\nabla_{y_\ell} u\|_2^2 \leq Cm, \quad \|W_{\text{ee}} \Pi\| \leq Cm^2. \quad (5.17)$$

The constants do not depend on the dimension $\binom{2m}{m}$ of $\text{Ran } \Pi$.

Proof. Let $(K_{\text{raw}})_{ij} = \langle \nabla \psi_i, \nabla \psi_j \rangle$. This is positive semidefinite, with trace at most Cm . In the orthonormal spatial modes the kinetic matrix is

$$K_0 = G^{-1/2} K_{\text{raw}} G^{-1/2}.$$

Hence

$$\text{Tr } K_0 = \text{Tr}(K_{\text{raw}} G^{-1}) \leq \|G^{-1}\| \text{Tr } K_{\text{raw}} \leq Cm.$$

The spin matrix $K = K_0 \otimes I_2$ has the same bound after adjusting the constant.

For any normalized fermionic u , its one-particle density matrix γ in these modes satisfies $0 \leq \gamma \leq I$. Indeed, for a coefficient vector v , the quadratic form of γ is $\|c(v)u\|^2$; the anticommutation relation $c(v)c(v)^\dagger + c(v)^\dagger c(v) = \|v\|^2 I$ places it between zero and $\|v\|^2$. Thus, for an arbitrary superposition in the finite-mode space,

$$\sum_{\ell} \|\nabla_{y_\ell} u\|_2^2 = \text{Tr}(\gamma K) \leq \text{Tr } K \leq Cm.$$

For a pair $\ell < k$, applying the three-dimensional Hardy inequality in y_ℓ , with y_k and all other variables held fixed, gives

$$\| |y_\ell - y_k|^{-1} u \|_2 \leq 2 \|\nabla_{y_\ell} u\|_2.$$

Integration over the remaining variables and spin components justifies this inequality for the full wavefunction. Put $b_m = m(m-1)/2$. Cauchy–Schwarz in the sum of pair multipliers gives, when $m \geq 2$,

$$\begin{aligned} \|W_{\text{ee}} u\|_2^2 &\leq b_m \sum_{\ell < k} \| |y_\ell - y_k|^{-1} u \|_2^2 \\ &\leq 4b_m \sum_{\ell < k} \|\nabla_{y_\ell} u\|_2^2 \\ &\leq 4b_m(m-1) \sum_{\ell} \|\nabla_{y_\ell} u\|_2^2 \leq Cm^4. \end{aligned}$$

For $m = 1$ the repulsion is zero. This proves the operator bound on the whole finite-mode space, rather than only on its individual Slater determinants. $\square$

Lemma 5.4 (Eliminating the many-particle complement). *For sufficiently large D , let $E_\Pi = \min \text{Spec}(\Pi \mathcal{H} \Pi|_{\text{Ran } \Pi})$. Then*

$$0 \leq E_\Pi - \inf \text{Spec } \mathcal{H} \leq e^{-1.2D}. \quad (5.18)$$

More explicitly, the upper bound can be taken to be

$$C \left(m^{3/2} e^{-.75D} + m^2 / Z_* \right)^2.$$

Proof. Let $\varepsilon = \|AP\| \leq C\sqrt{m}e^{-.75D}$. The one-particle off-block operator

$$R_{\text{off}} = PA(I - P) + (I - P)AP$$

extends to a bounded self-adjoint operator, of norm $\|(I - P)AP\| \leq \varepsilon$. The adjoint identifies the two blocks even though A itself is unbounded. Its block-diagonal remainder satisfies

$$A - R_{\text{off}} = PAP + (I - P)A(I - P) \geq -\varepsilon P + g(I - P)$$

by Proposition 5.2. Consequently, on the full many-particle form domain,

$$d\Gamma(A) \geq gN_{\perp} - 2m\varepsilon I, \quad N_{\perp} = d\Gamma(I - P). \quad (5.19)$$

Here the low diagonal can cost at most $m\varepsilon$, and $\|d\Gamma(R_{\text{off}})\| \leq m\varepsilon$. The commuting one-particle projections $(I - P)^{(\ell)}$ show that $N_{\perp}$ has nonnegative integer eigenvalues, with kernel exactly $\text{Ran } \Pi$. Therefore $N_{\perp} \geq Q_m$, including on superpositions of states with different numbers of high-mode particles.

The repulsion form is nonnegative on every wavefunction, so its compression to Q_m is nonnegative even though W_{ee} need not commute with Π . Equation (5.19) gives

$$Q_m \mathcal{H} Q_m \geq (g - 2m\varepsilon)Q_m \geq cQ_m, \quad c = g/2, \quad (5.20)$$

for large D .

By Lemma 5.3, the coupling from the finite subspace has operator norm at most

$$\beta := \|Q_m \mathcal{H} \Pi\| \leq m\varepsilon + \frac{Cm^2}{Z_*} \leq Cm^{3/2}e^{-.75D} + \frac{Cm^2}{Z_*}. \quad (5.21)$$

The first term follows because the block-diagonal one-particle sum preserves $\text{Ran } \Pi$. The compressed operator also obeys

$$\|\Pi \mathcal{H} \Pi\| \leq m\varepsilon + Cm^2/Z_* = o(1).$$

For any form-domain vector $u = p + v$, where $p = \Pi u$ and $v = Q_m u$, the finite-mode operator-domain inclusion and the bounded coupling justify

$$\begin{aligned} \langle u, \mathcal{H} u \rangle &\geq \langle p, \mathcal{H} p \rangle + c\|v\|^2 - 2\beta\|p\|\|v\| \\ &\geq \langle p, \mathcal{H} p \rangle - \frac{2\beta^2}{c}\|p\|^2 + \frac{c}{2}\|v\|^2. \end{aligned}$$

Equivalently, this is the form inequality

$$\mathcal{H} \geq \Pi \mathcal{H} \Pi - \frac{2\beta^2}{c}\Pi + \frac{c}{2}Q_m.$$

Since the finite block has norm $o(1)$, the bottom of the right side is $E_{\Pi} - 2\beta^2/c$ for large D . The opposite variational inequality follows by restricting test vectors to $\text{Ran } \Pi$. This proves the explicit error bound. In the present regime, with $Z_* \geq e^D/2$, it is at most $e^{-1.2D}$. All lower bounds apply to quadratic forms and hence to spectral infima; existence of an eigenvector at the bottom of the full continuum spectrum is not used. $\square$

In particular, the explicit error in this lemma, after multiplication by Z_* , is bounded by

$$C \left(m^3 e^{-.5D} + m^{7/2} e^{-.75D} + m^4 e^{-D} \right). \quad (5.22)$$

The weaker bound $Z_* e^{-1.2D} \leq 2e^{-.2D}$ will suffice. This squaring of the residual against the fixed complement gap is what permits the final energy amplification.

5.4 The compressed Coulomb interaction

The full-space energy is now controlled by its finite-mode compression, with an error that remains small after multiplication by Z_* . To identify that compression we retain the direct intersite repulsion and estimate it explicitly. At the present exponentially amplified scales it has norm $O(m^2/D)$ in Hubbard units, unlike constructions that must keep it at leading order at polynomial confinement scales.

Identify $\text{Ran } \Pi$ with $\bigwedge^m(\mathbb{C}^m \otimes \mathbb{C}^2)$ using the isometry $\mathcal{J}_m = \bigwedge^m \mathcal{J}$. Let

$$W_{\text{eff}} = \mathcal{J}_m^* W_{\text{ee}} \mathcal{J}_m.$$

Creation and annihilation operators below refer to these orthonormal abstract site modes.

Lemma 5.5 (Coulomb coefficients in the site modes). *For sufficiently large D ,*

$$\left\| W_{\text{eff}} - U \sum_{i=1}^m n_{i\uparrow} n_{i\downarrow} \right\| \leq \frac{Cm^5}{D}, \quad U = \frac{5}{8}. \quad (5.23)$$

Proof. For four spatial functions define

$$\mathcal{V}(a, b; c, d) = \iint \frac{\overline{a(x)b(y)} c(x)d(y)}{|x-y|} dx dy.$$

Hardy in the x coordinate and Cauchy–Schwarz show that

$$|\mathcal{V}(a, b; c, d)| \leq 2\|a\|_2 \|b\|_2 \|\nabla c\|_2 \|d\|_2. \quad (5.24)$$

Telescoping a change of the four arguments therefore makes this coefficient Lipschitz in their H^1 norms on uniformly bounded H^1 sets. The same statement holds for fixed spin orbitals, with the usual spin Kronecker factors.

The coefficients of W_{eff} are these integrals in the modes ω_i and their two spin copies. Equation (5.8) permits their replacement by ϕ_i with error at most Cm/D per coefficient. Indeed all four H^1 norms stay uniformly bounded. In the standard second-quantized expression each coefficient multiplies $\frac{1}{2}c_a^\dagger c_b^\dagger c_d c_c$, a monomial of norm at most one. There are at most $(2m)^4$ coefficients. Thus this replacement changes the operator by at most Cm^5/D . We are comparing coefficient arrays in the fixed abstract orthonormal site space; no orthogonality of the replacement functions ϕ_i is asserted or needed.

Choose a fixed smooth radial cutoff equal to one on $r_i \leq D/5$, zero on $r_i \geq D/4$, and with gradient bounded by C/D , and let ζ_i be its product with ϕ_i . Direct integration of the hydrogenic tail gives

$$\|\zeta_i - \phi_i\|_{H^1} \leq C \text{poly}(D) e^{-D/5} \leq e^{-.18D}.$$

The supports of the ζ_i are disjoint. By (5.24), replacing all ϕ_i by ζ_i changes the coefficient operator by at most $Cm^4 e^{-.18D}$.

For the truncated functions a nonzero coefficient must have the same site label for both factors at x , and likewise at y . Therefore the only surviving terms are onsite and intersite density terms. If $i \neq j$, their spatial coefficient satisfies

$$0 \leq V_{ij}^{\text{dir}} = \iint \frac{|\zeta_i(x)|^2 |\zeta_j(y)|^2}{|x-y|} dx dy \leq \frac{1}{D/2 - 2S} \leq \frac{C}{D},$$

using $\|\zeta_i\|_2 \leq 1$. The intersite operator is $\sum_{i<j} V_{ij}^{\text{dir}} n_i n_j$, where $n_i = n_{i\uparrow} + n_{i\downarrow}$ has norm two. It consequently has norm at most Cm^2/D .

The onsite spatial integral differs from $\mathcal{V}(\phi_i, \phi_i; \phi_i, \phi_i) = U$ by at most $Ce^{-.18D}$. The two equal-spin monomials vanish by anticommutation. The two opposite-spin monomials, together with the factor $1/2$ in the interaction, give exactly $Un_{i\uparrow}n_{i\downarrow}$. Combining these estimates yields

$$\left\| W_{\text{eff}} - U \sum_i n_{i\uparrow} n_{i\downarrow} \right\| \leq C \left(\frac{m^5}{D} + m^4 e^{-.18D} + \frac{m^2}{D} \right) \leq \frac{Cm^5}{D}.$$

□

5.5 The simulation estimate in physical energy units

Proposition 5.6 (Simulation of the Hubbard spectral minimum). *Let $E_0 = \inf \text{Spec } H$ for the physical molecular Hamiltonian and let $E_{\text{Hub}}(t)$ be the minimum of the half-filled Hubbard Hamiltonian in Proposition 2.6. The construction satisfies*

$$\left| \frac{E_0}{Z_*} + \frac{mZ_*}{2} - E_{\text{Hub}}(t) \right| \leq m\delta_1 + \frac{Cm^5}{D} + Z_* e^{-1.2D} \quad (5.25)$$

$$\leq C \left[m^2 \epsilon_{\text{hop}} + \frac{m^5}{D} + m e^{-.01D} + m^2 e^{-.1D} + m^2 e^{-.6D} + e^{-.2D} \right]. \quad (5.26)$$

In particular, for every prescribed fixed inverse-polynomial error budget ϵ_ , the polynomial scale and calibration precision can be chosen so that the left side is at most ϵ_* .*

Proof. The exact compression, in the abstract site space, is

$$\mathcal{J}_m^*(Z_* \mathcal{H}) \mathcal{J}_m = d\Gamma(Z_* \mathcal{J}^* A \mathcal{J}) + W_{\text{eff}}.$$

The Hubbard hopping term is $d\Gamma(\mathbb{T} \otimes I_2)$. On m particles, second quantization of a bounded one-particle difference increases its norm by at most a factor m . Lemma 5.1 and Lemma 5.5 therefore give

$$\|\mathcal{J}_m^*(Z_* \mathcal{H}) \mathcal{J}_m - H_{\text{Hub}}(t)\| \leq m\delta_1 + Cm^5/D.$$

For finite self-adjoint matrices, the difference of their spectral minima is bounded by the operator norm difference, by the variational principle. Lemma 5.4 adds at most $Z_* e^{-1.2D}$ when the restriction to $\text{Ran } \Pi$ is removed. Finally, (5.16) gives the exact energy identity

$$Z_* \inf \text{Spec } \mathcal{H} = \frac{E_0}{Z_*} + \frac{mZ_*}{2}.$$

This proves (5.25). Substitute (5.7) and $Z_* \leq 2e^D$ to obtain (5.26).

The right side involves only fixed polynomial factors in m , the calibrated hopping error, inverse powers of D , and the displayed decaying exponentials. Equation (5.5) and Proposition 4.8 allow the hopping error to be made an arbitrarily small fixed inverse polynomial. The simultaneous choices, their order, and polynomial bit cost are specified in Section 6. No selected-orbital assumption has been imposed on the spectral infimum in this comparison. □

6 Parameters, computation, and the final promise

We now put the constructions in their algorithmic order. This matters because the tail formulas are proved for tuned wells, whereas the positions must be computed before the tuning is performed. The surrogate fields in [Section 4](#) remove this apparent dependence.

6.1 A uniform choice of scales

Let $n \geq 2$ bound the source input size, with the fixed polynomial magnitude and inverse-gap bounds from [Theorem 2.1](#). Write $E_{\text{in}} = E(H_{\text{in}})$. Put

$$\delta_s = b_s - a_s, \quad \gamma = \min\{1, \delta_s\}, \quad \alpha = \eta^2/U.$$

Use the scales and rational hopping approximation of [Theorem 2.6](#). They give

$$|E_{\text{Hub}}(t) - \alpha(E_{\text{in}} - C_s)| \leq \frac{3\alpha\gamma}{64}, \quad C_s = 3k\Delta + \frac{3}{2} \sum_e v_e^2 + \sum_{\{i,j\} \in \mathcal{C}} K_{ij}. \quad (6.1)$$

All parameters chosen so far have polynomial magnitude or inverse-polynomial lower bounds as appropriate. Their binary descriptions have polynomial length. The scale α is rational. The scalar C_s is a sum of polynomially many efficiently computable square roots and rational terms; it does not depend on E_{in} .

Set the requested continuum error to

$$\epsilon_* = \frac{\alpha\gamma}{64}. \quad (6.2)$$

We verify that a single deterministic polynomial choice of D meets this budget. The calibrated displacement bound has the form

$$S \leq Cn^b(1 + \log D) \quad (6.3)$$

for a fixed b , since m , the number of contacts, and $|\log t_{ij}|$ are polynomially bounded in n . Increase b so that $m \leq Cn^b$ as well. The hopping error before polynomial norm conversions is bounded by a constant times

$$\begin{aligned} \mathfrak{e}(D, \tau) &= \frac{m \log D + m^2}{D} + \frac{mS}{D} + \frac{S^2}{D} + SD^{-1/4} \\ &\quad + \tau + e^{-c\sqrt{D}}. \end{aligned} \quad (6.4)$$

The other errors are $\text{poly}(m)/D$ and the exponentially small terms listed below. Choose fixed exponents h, r so that all polynomial norm-conversion and coefficient-counting losses are bounded by Cn^h , and $\epsilon_* \geq cn^{-r}$. In particular, take $m^2 + m^5 \leq Cn^h$; the two algebraic error terms in [Equation \(5.26\)](#) are then bounded by $Cn^h(\mathfrak{e}(D, \tau) + D^{-1})$. Such exponents exist by the explicit finite construction. Increasing them only weakens the conditions to be met.

Choose an integer K satisfying

$$K > 4(b + h + r + 2), \quad (6.5)$$

and put

$$D = A(n + 2)^K, \quad \tau = c_*(n + 2)^{-h-r-2}, \quad (6.6)$$

where A is a sufficiently large fixed integer and $c_* > 0$ a sufficiently small fixed rational. The factors $1 + \log D$ in [Equation \(6.3\)](#) do not make this choice circular: K is chosen first, and then A

is fixed. Each D -dependent expression tends to zero relative to its error budget as A increases, even with its powers of $\log A$. The remaining growth in n is controlled by Equation (6.5). In particular,

$$n^h S D^{-1/4} = o(n^{-r}), \quad n^h S^2 / D = o(n^{-r}), \quad m + S = o(D^{1/4}).$$

Here the little-oh statements describe the uniform large-scale estimates used to choose the fixed constants; increasing A covers every finite small- n case. The same choices bound all the terms in Equation (6.4) and the Coulomb-compression error by their assigned fractions of Equation (6.2). No search for a successful D is part of the algorithm.

It is important to keep the energy units explicit. After dilation and subtraction of the isolated-well ground energy,

$$\mathcal{H} = H/Z_*^2 + m/2, \quad Z_* \mathcal{H} = H/Z_* + mZ_*/2. \quad (6.7)$$

The exponential margins that are needed after multiplication by $Z_* \asymp e^D$ are:

Source of error	In $\mathcal{H}$ units	In $Z_* \mathcal{H}$ units
Tuned well energies	$m e^{-1.01D}$	$O(m e^{-.01D})$
Discarded one-body entries	$O(m^2 e^{-1.1D})$	$O(m^2 e^{-.1D})$
Orthonormalization	$\text{poly}(m) e^{-1.6D}$	$\text{poly}(m) e^{-.6D}$
Continuum complement	$e^{-1.2D}$	$O(e^{-.2D})$

The polynomial multiplicities are retained in Equation (5.26). The positive exponential margins allow all these terms to fit their assigned error budgets by increasing the fixed A . The one-body residual $e^{-.75D}$ is used quadratically against a fixed complementary gap; multiplying that residual linearly by Z_* would not be a valid estimate. Theorem 5.6 supplies the required quadratic bound.

With these choices, that proposition gives

$$|E_0/Z_* + mZ_*/2 - \alpha(E_{\text{in}} - C_s)| \leq \frac{\alpha\gamma}{16} \leq \frac{\alpha\delta_s}{16}. \quad (6.8)$$

6.2 The deterministic construction and its bit cost

First check the source encoding and the fixed polynomial bounds in Theorem 2.1, including the inverse-gap bound. If a check fails, output the fixed instance with one electron, one unit-charge nucleus at the origin, and thresholds $-1/2$ and $1/2$. Otherwise carry out the following steps. The guard is polynomial-time and makes the reduction a total polynomial-time map; it never rejects a promised source instance.

1. *Finite models.* Read the signed rational Heisenberg instance. Form the ancillary graph, the algebraic reference positions p_i , the complete contact set $\mathcal{C}$, the explicit scales Δ, η , and the positive rational hoppings t_{ij} of Theorem 2.6. The finite contact classification avoids deciding arbitrary geometric equalities numerically. All square roots here are approximated with certified rational intervals to the stated inverse-polynomial errors.
2. *Charge scale.* Choose D as in Equation (6.6) and compute a positive integer Z_* with

$$e^D/2 \leq Z_* \leq 2e^D.$$

This does not require an exact comparison with a transcendental number. For example, compute the rational sum $s_D = \sum_{j=0}^{4D} D^j/j!$ and take $Z_* = \lfloor s_D \rfloor$. The ratio of successive omitted terms

is at most $1/4$, and $j! \geq (j/e)^j$, so $0 < e^D - s_D < 1$ for sufficiently large D . Thus this choice has the required bounds. The sum has polynomially many terms and polynomial bit cost. Round $Z_*(1 - 500m/D)$ and Z_*1000m/D to the positive integer primary and secondary charges. They are fixed before any energy tuning.

3. *Positions.* At the reference arrangement $X_i^0 = Dp_i$, compute each positive surrogate hopping integral to the required relative accuracy, then compute

$$d_{ij} = \log(Z_*T_{ij}(X^0)/t_{ij})$$

to the prescribed inverse-polynomial error. Apply the explicit right inverse from [Theorem 2.5](#) and round the resulting scaled centers to rational X_i . The final displacements are defined by $s_i = X_i - Dp_i$; thus the exact contact-vector decomposition is retained after rounding. [Theorem 4.8](#) proves that these operations take polynomial time and satisfy the required projected-displacement tolerances. The calculation uses W_i^* and is independent of the unknown tuned parameters.

4. *Well tuning.* With these final rational centers and integer charges fixed, apply the certified one-well oracle and the contraction iteration of [Theorem 3.7](#). This produces rational $\rho_i \in [1, 4]$ such that $|\lambda_i + 1/2| \leq e^{-1.01D}$. No subsequent change to a charge or center is made.
5. *Physical instance.* Output the $2m$ nuclei at

$$R_{i,0} = X_i/Z_*, \quad R_{i,1} = (X_i + \rho_i e_z)/Z_*$$

with their positive integer charges, and output $N = m$ in unary. Compute the rational thresholds specified in the next subsection.

Each step is classical and deterministic. The one-well Ritz spaces have dimension polynomial in D and in the rational coordinate bit lengths. Their matrix entries and conditioning bounds require only polynomially many bits, and the number of eigenvalue evaluations in the tuning procedure is polynomial. The integral evaluations and elementary functions in calibration likewise require only polynomial time and precision. In particular, no many-electron ground energy is queried.

The output has polynomial length. Each nuclear charge has $O(D + \log n)$ bits. Rational scaled centers and secondary offsets have polynomial bit length, and division by the $O(D)$ -bit integer Z_* preserves that property. Distinct sites are separated by $D - O(S)$ in scaled coordinates; each secondary nucleus is between one and four units from its own primary. Thus all output nuclei are distinct for the chosen D . Reducing rational coordinates and thresholds to lowest terms is a polynomial-time integer computation. Finally, m is polynomial in n , so unary encoding of N has polynomial length as well. Auxiliary caps, orbitals, Ritz bases, and projections are used in the proof and computation only; none is part of the output instance.

6.3 Rational thresholds and preservation of both promises

Compute a rational $\tilde{C}_s$ with

$$|\tilde{C}_s - C_s| \leq \delta_s/64. \tag{6.9}$$

Polynomially many square-root approximations suffice, since C_s has polynomial magnitude and δ_s has an inverse-polynomial lower bound. Define the rational affine map

$$\tilde{F}(z) = -mZ_*^2/2 + Z_*\alpha(z - \tilde{C}_s), \tag{6.10}$$

and set

$$a = \tilde{F}(a_s + \delta_s/4), \quad b = \tilde{F}(b_s - \delta_s/4). \quad (6.11)$$

All the quantities in these formulas are rational and computable with polynomial bit cost. The possibly enormous common offset $-mZ_*^2/2$ is an exact rational with polynomial bit length.

By [Equations \(6.8\) and \(6.9\)](#),

$$|E_0 - \tilde{F}(E_{\text{in}})| \leq \frac{5}{64} Z_* \alpha \delta_s < \frac{1}{8} Z_* \alpha \delta_s. \quad (6.12)$$

Since $\tilde{F}$ has positive slope, a YES source instance satisfies

$$E_0 \leq \tilde{F}(a_s) + \frac{1}{8} Z_* \alpha \delta_s \leq a - \frac{1}{8} Z_* \alpha \delta_s < a,$$

and a NO source instance satisfies

$$E_0 \geq \tilde{F}(b_s) - \frac{1}{8} Z_* \alpha \delta_s \geq b + \frac{1}{8} Z_* \alpha \delta_s > b.$$

These inequalities concern the exact continuum spectral infimum by [Theorem 5.6](#), rather than only a projected eigenvalue.

The threshold separation is exactly

$$b - a = \frac{1}{2} Z_* \alpha \delta_s. \quad (6.13)$$

The reciprocal of $\alpha \delta_s$ is polynomially bounded in n , whereas $Z_* \geq e^D/2$. Increasing the same fixed scale constant if necessary makes [Equation \(6.13\)](#) at least one for every input. Since the output length $L \geq 1$, this implies $b - a \geq L^{-1}$ without any circular estimate involving L .

We have constructed a legal nuclear instance in deterministic polynomial time, preserving YES and NO and permitting all spin sectors and all continuum states. This proves [Theorem 1.1](#).

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